Machine-Checked Dimension Theory for Fusion Categories

A research proposal: pivotal and spherical structures, the double-dual twist, and a verified proof of Etingof–Nikshych–Ostrik Proposition 8.23

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Abstract

This proposal sets out a research programme whose central target is the first machine-checked proof of Etingof–Nikshych–Ostrik Proposition 8.23: *every pseudo-unitary fusion category admits a unique spherical structure whose dimensions are the Frobenius–Perron dimensions.* The theorem is the standard bridge from an analytic hypothesis (one scalar equation on the global dimension) to the algebraic structure that all of dimension theory presupposes, and it is used as a black box throughout the classification literature. No formal proof of it exists, and the categorical infrastructure on which it rests is only now arriving in Mathlib.

The programme builds on substantial preliminary work, reported in §4: a skeletal formalisation of multiplicity-free rigid fusion data with a proved inhabitant; a corrected duality normalisation that inhabits Fibonacci and Ising over $\mathbb{R}$; a reduction of the target to a single per-channel identity on the double-dual twist; a formalised proof of ENO Proposition 8.21; and a reconstruction of ENO's route through the square root a of the quadruple-dual isomorphism, together with the identification of ENO's trace identity with multiplicativity of the dimension. One of these results is negative and changes the shape of the target: ENO 8.23 is *false* without its pseudo-unitarity hypothesis (a kernel-checked two-simple counterexample, the Yang–Lee category). A second witness shows that the dimension function depends on the choice of pivotal structure although the fusion ring is unchanged. The proposal turns these findings into a three-aim programme: a corrected and inhabitable formal signature together with its bridge to Mathlib's emerging pivotal and spherical category theory; a derivation of the double-dual twist from the duality data; and a formalisation of ENO's analytic core, yielding Proposition 8.23 in the scope actually formalised — multiplicity-free data, with the analytic theorem stated over $\mathbb{C}$ (FP dimensions embedded from $\mathbb{R}$, positivity defined once), and the quadruple-dual theorem either taken as a hypothesis or proved by Bartlett's string-diagram argument.

The scope is narrower than the title of the source theorem and is fixed in §2.6: multiplicity-free data; the algebraic layer over a field of characteristic zero with the analytic theorem over $\mathbb{C}$; and the quadruple-dual isomorphism either as a hypothesis or proved by Bartlett's string-diagram argument. The 24-month schedule is ordered by risk — corrected signature and exact target, reconstruction and trace correspondence, derived twist, ENO's normalisation — and the deliverables include an axiom-clean Lean 4 library, a verified negative-result note, upstream Mathlib contributions, and a documented methodology for auditing formal statements against their sources.

At a glance.	
Object	Pivotal and spherical structures on fusion categories, and ENO Prop. 8.23.
Method	Skeletal finite data in Lean 4 (pinned to Mathlib `0df444a3`); exact arithmetic in $\mathbb{Z}[\sqrt{d}]$; kernel-checked counterexamples; axiom audit; bridge to Mathlib's categorical layer.
Already done	∼8,100 lines, 31 modules; a reproduced clean build (batch B1: fresh clone, fresh Mathlib cache, no path override; content-reproducible, not byte-reproducible); 1,058 audited declarations at `[propext, Classical.choice, Quot.sound]`, mechanically enumerated; no `sorry`.
Still open	A corrected signature and an exact target; the twist derivation (the Ω formula in `PLAN.md` is retracted); the $\mathbb{C}$ scalar contract; the reconstruction and trace correspondence; ENO's σ/τ normalisation; the quadruple-dual theorem.
Horizon	24 months, 8 work packages (WP0–WP7), 12 numbered deliverables; a first delegation batch (B1–B4, §5.2) gates all dependent theorem work.

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1 Introduction

1.1 Dimension theory sits between decidable combinatorics and open structure

A fusion category over a field $\Bbbk$ is a $\Bbbk$-linear, semisimple, rigid monoidal category with finitely many simple objects and $\mathrm{End}(\mathbf{1}) = \Bbbk$. Its Grothendieck ring $K_0(\mathcal{C})$ is finite, decidable data: a finite set I of simples, integers $N_{ij}^k = \dim \mathrm{Hom}(X_k, X_i \otimes X_j)$, a duality involution $i \mapsto i^*$. Almost everything one wants to compute about $\mathcal{C}$ — dimensions, the global dimension, Frobenius–Schur indicators, the T-matrix of a modular category, the state sum of a three-manifold — depends on more than this ring, but less than the full category.

The intermediate datum is a *pivotal structure*: a monoidal isomorphism $\gamma\colon \mathrm{Id} \to (-)^{**}$. It yields a dimension $\dim_\gamma(X) = \mathrm{tr}(\gamma_X)$ for every object, and a *spherical* structure — one with $\dim_\gamma(X) = \dim_\gamma(X^*)$ — is what makes the trace cyclic and therefore produces invariants of oriented manifolds rather than of framed ones.

Two facts locate the research problem precisely.

(a) *The dimension function is not a function of the fusion ring.* This is not a philosophical remark: the present project contains a kernel-checked witness (§4, `VecZ2Twisted`). The category $\mathrm{Vec}^{\omega}_{\mathbb{Z}/2}$ with non-trivial cocycle carries two pivotal structures whose dimensions on the non-trivial simple are $+1$ and -1 respectively. Any formalisation that treats "the dimension" as determined by the fusion rules is therefore wrong, and the first version of this project's pivotal axiom was wrong in exactly that way.

(b) *Whether a pivotal structure exists is open in general.* Etingof–Nikshych–Ostrik conjectured that every fusion category is pivotal (ENO, Conjecture 2.8); the spherical strengthening is stronger still. What is known is a sufficient condition of a different character: pseudo-unitarity.

1.2 The theorem that bridges the two

ENO Proposition 8.23 states that a pseudo-unitary fusion category — one whose categorical global dimension equals its Frobenius–Perron global dimension — admits a *unique* pivotal structure, which is spherical, and with respect to which the categorical dimensions of all simple objects are positive and equal to their Frobenius–Perron dimensions.

The proposition is used as a black box across the classification programme: in Ostrik's analysis of pivotal fusion categories of rank 3 and the resulting inequality, in the Haagerup–Izumi existence problem, in the passage from unitary anyon data to modular data, and in every place where one writes down a spherical category from a fusion ring plus positivity. It is, in a precise sense, the lemma that makes the rest of the subject work.

1.3 Why formalise, and why this is not transcription

A referee reading a two-page proof in a 2005 Annals paper cannot check, line by line, that the gauge chosen for the associator is the one in which the evaluation and coevaluation have the normalisation the argument assumes. A proof assistant can. The empirical case for this proposal is that when the present project first wrote down the statement, four defects surfaced, every one of them invisible to a reader checking that the formulas *looked* right:

D1. The signature had no inhabitants. The two unit axioms of the skeletal fusion datum were mutually inconsistent for $n \geq 2$: evaluated at $(0, b, 0, d, b, 0)$ they forced $1 = 0$. Every theorem quantified over the structure was therefore vacuous, including a "proof" of the target theorem.

D2. The pentagon was not the pentagon. The axiom called `pentagon` had eight free labels where the F-matrix pentagon requires ten, and it fails on $\mathrm{Vec}^{\omega}_{\mathbb{Z}/2}$ for *every* cocycle ω, the trivial one included. A green build certified a false axiom.

D3. The pivotal axiom fixed the twist to one. `p_fusion` assumed $J \equiv 1$, where J is the monoidal structure of $(-)^{**}$. With that assumption every fusion-rule table is pivotal via $p \equiv 1$, and the target theorem became provable with no hypothesis at all — the formal statement had drifted so far from the source that it asserted something false.

D4. The second snake used the wrong inverse. It wrote the right snake with a reciprocal of an associator *entry* where the $(0,0)$-entry of the *matrix inverse* of the associator block is required. The two agree for pointed categories and differ for Fibonacci, so the axiom excluded the two principal examples.

Each defect was found by making the statement inhabitable and checking it against the examples, not by re-reading it. This is the methodological thesis of the proposal: in this subject the value of formalisation lies less in the final certificate than in the audit that produces it, and the audit is transferable.

1.4 What has already been established

The preliminary results of §4 are not a plan; they are a library. In summary:

- a skeletal signature for multiplicity-free rigid fusion data, with a proved inhabitant ($\mathrm{Vec}_{\mathbb{Z}/2}$) and the pentagon checked on all 512 label tuples;
- a corrected duality normalisation, with the inverse associator carried as data, that inhabits Fibonacci and Ising over $\mathbb{R}$ — every field proved, including both snakes — and yields the spherical pivotal structure with dim = FPdim for both;
- a proof that a pivotal dimension is always a coboundary of the normalised double-dual twist, and that ENO 8.23 is *equivalent* to one per-channel identity on that twist;
- a formalised ENO Proposition 8.21: the spectral bound $|d_i| \leq \mathrm{FPdim}_i$ for any dimension function satisfying the eigenvector equation, together with the equality case;
- a kernel-checked refutation of the target as stated without hypotheses (Yang–Lee), and a second witness that the dimension function is not determined by the fusion ring;
- a reconstruction of ENO's own proof route — the square root a of the quadruple-dual isomorphism, its monoidality defect b, and the trace identity — together with a proof that ENO's trace identity *is* the multiplicativity identity that the twist reduction isolates.

The library builds (3,032 jobs, warnings only), contains no `sorry`, and its axiom audit over 73 declarations reports `[propext, Classical.choice, Quot.sound]` throughout.

1.5 What is not yet established

The gap is now sharp, and it is not the algebra. It is ENO's analytic argument: the passage from the pseudo-unitarity equation (one scalar identity, Definition 2.7) to a dimension function that is positive and multiplicative. Proposition 8.21 turns the *sum* equality $\sum_i |d_i|^2 = \sum_i \mathrm{FPdim}_i^2$ into a termwise equality $|d_i| = \mathrm{FPdim}_i$; a tensor automorphism of the identity, $\sigma = d^2/|d|^2$, then makes the dimensions real, and a *natural* automorphism $\tau = d/|d|$ — not monoidal in general — makes them positive. No Hermitian structure enters the hypothesis or the proof: positivity is part of what the argument *produces*, and an earlier draft of this section described the gap as a passage from a Hermitian structure; that description is removed here and in the plan. Alongside it, and logically prior to it, sit two defects of the signature rather than of the mathematics: the twist J is carried as data constrained by four laws, not derived from $(\mathrm{ev}, \mathrm{coev}, F)$, and the scalars are real where the source theorem is over $\mathcal{C}$. Until J is derived, the statement "p is pivotal" mentions an object the mathematics does not fix.

The scope is narrower than the source theorem and is fixed here once (§2.6): multiplicity-free data; the algebraic layer over a field of characteristic zero with the analytic theorem over $\mathbb{C}$ (S2 of §2.6); and the quadruple-dual isomorphism g of ENO Theorem 2.6 either taken as a hypothesis or

proved by the string-diagram argument of Bartlett §3.6. These are deliverables to be discharged, not silent assumptions.

1.6 Research questions

Q1 (Signature).
What is the weakest finite-data signature that is (i) inhabited by the standard examples, (ii) strong enough to express pivotality and sphericity faithfully, and (iii) provably equivalent to the abstract categorical notion?

Q2 (Twist).
Is the double-dual twist J determined by $(F, \mathrm{ev}, \mathrm{coev})$? If so, what is its closed form, and does it satisfy the multiplicativity identity (M)?

Q3 (Analytic core).
Can ENO's proof of Proposition 8.23 be carried out inside the skeleton, taking as input only the pseudo-unitarity equation $\dim \mathcal{C} = \mathrm{FPdim}\mathcal{C}$?

Q4 (Frontier).
What is the largest statement provable *without* pseudo-unitarity — that is, what exactly is the content of ENO Conjecture 2.8 in a pivotal-free formalism, and which of its consequences survive as theorems about torsion in the universal grading group?

Q5 (Method).
Which failure modes of formalising monoidal category theory are systematic, and can they be detected mechanically rather than by luck?

1.7 Structure of the proposal

§2 fixes notation and states the target precisely. §3 surveys prior work, mathematical and formal. §4 reports the preliminary results in enough detail to be checked. §5 sets out the eight work packages and their milestones. §6 describes the techniques, including the ones that failed. §7–§9 give risks, timeline and deliverables. §10 argues for significance. The appendices record the module inventory, the audit protocol, the Lean statements of the main targets, and the changes made in each revision.

2 Background

2.1 Fusion categories and fusion rings

Definition 2.1. *A* fusion category *over a field $\Bbbk$ is a $\Bbbk$-linear abelian, semisimple, rigid monoidal category $\mathcal{C}$ with finitely many isomorphism classes of simple objects, such that* $\mathrm{End}_{\mathcal{C}}(\mathbf{1}) \simeq \Bbbk$ *and the tensor product is bilinear and exact in each variable. We write I for the set of isomorphism classes of simples, X_i for a representative, $0 \in I$ for the class of $\mathbf{1}$, and $i \mapsto i^*$ for the duality involution determined by* $\mathrm{Hom}(X_i \otimes X_j, \mathbf{1}) \neq 0 \iff j = i^*$.

The Grothendieck ring $K_0(\mathcal{C})$ has $\mathbb{Z}$-basis (X_i) and structure constants

$$X_i X_j = \sum_{k \in I} N_{ij}^k X_k, \qquad N_{ij}^k = \dim_{\Bbbk} \mathrm{Hom}(X_k, X_i \otimes X_j) \in \mathbb{N}.$$

In this proposal a *fusion ring* is the resulting abstract datum $R = (N, (-)^*, 0)$: the multiplicities, the duality involution, and the unit. The axioms used are those of a based ring: $N_{ij}^0 = \delta_{j,i^*}$; associativity $\sum_k N_{ij}^k N_{kl}^m = \sum_k N_{jl}^k N_{ik}^m$; and compatibility of the involution with the product. Commutativity is *not* assumed.

Definition 2.2. *A fusion ring is* multiplicity-free *if* $N_{ij}^k \in \{0,1\}$ *for all* i,j,k. *It is* pointed *if every simple object is invertible, i.e.* $N_{ij}^k \in \{0,1\}$ *and* $\sum_k N_{ij}^k = 1$ *for all* i,j.

Multiplicity-freeness is not a convenience in this project: with $N > 1$ the associator is a matrix between multi-dimensional Hom spaces, and a single scalar $F_d^{abc}(e,f)$ cannot hold it. The restriction is discussed in §6.4.

2.2 The Frobenius–Perron dimension

Theorem 2.3 (Frobenius–Perron, for fusion rings)**.** *Let* R *be a fusion ring with* n *classes. There is a unique function* $f : I \to \mathbb{R}_{>0}$ *with* $f_0 = 1$ *and*

$$f_i f_j = \sum_k N_{ij}^k f_k \qquad \text{for all } i,j \in I. \tag{1}$$

Moreover $f_{i^*} = f_i$, *each* f_i *is an algebraic integer, and if the fusion graph is connected then* f *is the Perron–Frobenius eigenvector of every fusion matrix.*

The number f_i is the *Frobenius–Perron dimension* $\mathrm{FPdim}(X_i)$, and $\mathrm{FPdim}(\mathcal{C}) = \sum_i f_i^2$ is the Frobenius–Perron global dimension. Note that (1) is the *character* form: f is a ring homomorphism $K_0(\mathcal{C}) \to \mathbb{R}$.

In the present library Theorem 2.3 is not proved: the existence and uniqueness of f is packaged as a structure `FPData`, supplied by hand for each example. Discharging it from Mathlib's Perron–Frobenius development is a deliverable (§5, WP0), not a research risk: the relevant material exists upstream as a chain of pull requests (§3).

2.3 Duality, pivotal structures, and the double-dual twist

Rigidity supplies, for each object X, an evaluation $\mathrm{ev}_X : X^* \otimes X \to \mathbf{1}$ and a coevaluation $\mathrm{coev}_X : \mathbf{1} \to X \otimes X^*$ satisfying the two snake equations. In a skeleton, both are scalars, and the composite

$$\kappa_i \;=\; \mathrm{ev}_i \cdot \mathrm{coev}_i \;\in\; \Bbbk^\times$$

is the loop scalar of X_i. It is *not* the dimension: the dimension depends on a choice of identification of X with its double dual.

Definition 2.4. *A* pivotal structure *on* $\mathcal{C}$ *is a monoidal natural isomorphism* $\gamma : \mathrm{Id}_{\mathcal{C}} \to (-)^{**}$. *Its* dimension *is* $\dim_\gamma(X) = \mathrm{tr}(\gamma_X : X \to X^{**})$, *computed as the composite of* γ_X *with the evaluation of the right dual. The structure is* spherical *if* $\dim_\gamma(X) = \dim_\gamma(X^*)$ *for every* X.

Two remarks are essential and both cost the project a rewrite.

Remark 2.5 (The twist). The functor $(-)^{**}$ is monoidal, and its monoidal structure is *not* the identity. Write J_{ij}^k for the scalar of $X_i^{**} \otimes X_j^{**} \to (X_i \otimes X_j)^{**}$ on the channel k. Then a pivotal structure on a skeleton is exactly a family $(p_i)_{i \in I} \subset \Bbbk^\times$ with

$$p_0 = 1, \qquad p_k = J_{ij}^k \, p_i \, p_j \quad \text{whenever } N_{ij}^k > 0, \tag{2}$$

and the resulting dimension is

$$\dim_i \;=\; p_i \, \kappa_i. \tag{3}$$

Sphericity becomes

$$\dim_{i^*} = \dim_i \;\; \forall i \qquad \Longleftrightarrow \qquad \kappa_{i^*} = p_i^2 \, \kappa_i \;\; \forall i. \tag{4}$$

Remark 2.6 (The trap). Two statements are easily conflated, and the project conflated them once. (i) *Objectwise*, every simple object V of a fusion category is isomorphic to its double dual V^{**} (ENO Proposition 2.1): the right dual V^* is the unique simple X with $V \otimes X \ni \mathbf{1}$, the left dual ${}^*V$ is characterised the same way, and semisimplicity gives ${}^*V = V^*$, whence $V \simeq V^{**}$.

Consequently a skeleton may always be chosen with $i^{**} = i$ on the nose, and the signature does not restrict generality by assuming it. (ii) *Monoidally*, the functor $(-)^{**}$ need not be isomorphic to the identity, and a monoidal such isomorphism is *precisely* a pivotal structure. Statement (i) fixes the objects and says nothing about the monoidal structure; that structure is the twist J, and it is why J is data. An earlier version of this project built (ii) into the signature as $J \equiv 1$, which made every fusion-rule table pivotal and the target theorem true without hypotheses; the claim that a skeleton can always be chosen with $(-)^{**} = \mathrm{id}$ *monoidally* is false, and it is retracted in `PLAN.md`.

2.4 Pseudo-unitarity and the target

Definition 2.7 (Squared norms; ENO §2.1, §8.4; EGNO Definition 9.4.4)**.** *For a simple V and any isomorphism $a\colon V \to V^{**}$ (which exists by ENO Proposition 2.1), the* squared norm *is*

$$|V|^2 \;=\; \mathrm{Tr}_V(a)\,\mathrm{Tr}_{V^*}\big((a^{-1})^*\big) \;\in\; \Bbbk^\times,$$

independent of a. The global dimension *is* $\dim \mathcal{C} = \sum_i |X_i|^2$*, and $\mathcal{C}$ is* pseudo-unitary *if* $\dim \mathcal{C} = \mathrm{FPdim}\mathcal{C} = \sum_i f_i^2$.

Definition 2.7 is the non-circular one: it uses only the objectwise isomorphisms of ENO Proposition 2.1, not a pivotal structure, and is therefore available *before* the existence theorem that is to be proved. In a skeleton with a chosen a_i, one has $|X_i|^2 = \kappa_i \kappa_{i^*}$; the library proves the form-independence of this product as `dim_mul_dim_dual`, so pseudo-unitarity is the equation

$$\sum_{i \in I} \kappa_i\, \kappa_{i^*} \;=\; \sum_{i \in I} f_i^2 \tag{5}$$

with no pivotal structure in it. The library's `IsPseudoUnitary` is exactly (5).

Theorem 2.8 (ENO Proposition 8.23; EGNO Proposition 9.5.1)**.** *Let $\mathcal{C}$ be a pseudo-unitary fusion category. Then $\mathcal{C}$ admits a pivotal structure γ; γ is spherical;* $\dim_\gamma(X) > 0$ *for every $X \neq 0$;* $\dim_\gamma(X_i) = \mathrm{FPdim}(X_i)$ *for every simple X_i; and γ is the unique such structure.*

In this proposal the theorem is read over $\mathbb{C}$: "$\dim_\gamma(X) > 0$" through the positivity predicate of S2, the FP dimensions embedded from $\mathbb{R}$; the $\mathbb{R}$ form is the $k = \mathbb{R}$ specialisation, toward which the library currently works.

The two supporting results that ENO use are:

Theorem 2.9 (ENO Proposition 8.21)**.** *Let $d\colon I \to \mathbb{C}$ satisfy the eigenvector equation $\sum_j N_{ij}^k d_j = d_i d_k$ for all i, k. Then $|d_i| \leq \mathrm{FPdim}_i$ for all i. If moreover $\sum_i |d_i|^2 = \sum_i \mathrm{FPdim}_i^2$, then $|d_i| = \mathrm{FPdim}_i$ for all i.*

Theorem 2.10 (ENO Theorem 2.6, after Radford)**.** *For any fusion category there is a monoidal natural isomorphism $g\colon \mathrm{Id} \to (-)^{****}$.*

Theorem 2.10 is taken as a hypothesis in this project (§4): proving it requires the weak Hopf algebra machinery of ENO §4–5 and is explicitly out of scope. Conjecture 2.8 of ENO — that every fusion category admits a pivotal structure — is not attacked directly; WP4 in §5 explains what the programme can say about it.

2.5 The skeletal point of view and what it costs

The formalisation works with finite data over a field $\Bbbk$:

$$\big(N_{ij}^k,\; i^*,\; F_d^{abc}(e,f),\; F^{-1},\; \mathrm{ev}_i,\; \mathrm{coev}_i,\; J_{ij}^k\big)$$

subject to the pentagon, the unit normalisations, the two inverse laws for the associator, the two snake equations, and the twist laws. This is the "F-symbol" presentation familiar from the anyon-model literature, and it has three costs, each addressed by a work package.

(i) **No abstract coherence.** A skeletal datum is a category only after a reconstruction theorem. WP5 supplies the bridge to Mathlib's categorical definitions and deletes the vendored patch once the upstream pull requests land.

(ii) **Real scalars where the source is complex.** The examples are over $\mathbb{R}$ and $\mathbb{Q}$; the source theorem is over $\mathbb{C}$, where a pseudo-unitary datum typically produces roots of unity. WP1 Task 1.1 fixes the scalar contract: algebraic layer over a characteristic-zero field, analytic statements over $\mathbb{C}$ with the FP dimensions embedded from $\mathbb{R}$ and positivity defined once (S2 of §2.6). One restriction that looked like a cost is not one: by ENO Proposition 2.1 a skeleton always has $i^{**} = i$, so the signature does not restrict generality there (§2.3, Remark 2.6).

(iii) **Multiplicity-freeness.** The signature assumes $N \leq 1$ and the programme does not relax it: §6.4 explains why a matrix-valued associator is a separate development rather than a tweak, and scope statement (S1) keeps it out of the 24-month deliverable.

2.6 The scope, stated once

The title of ENO Proposition 8.23 is unconditional; what this proposal can deliver in 24 months is not, and the difference is stated here so that no later claim can quietly exceed it.

(S1) **Multiplicity-free data.** The signature carries a scalar associator, so $N_{ij}^k \leq 1$. The general case needs matrix-valued associators and is a separate development (§6.4); it is a stretch goal reached through the categorical bridge, not a consequence of the multiplicity-free theorem.

(S2) **The scalar contract: algebraic generality, analytic specificity.** The algebraic infrastructure — fusion rings, systems, pivotal structures, the twist reduction, the census — is developed over a field $\Bbbk$ of characteristic zero, where every statement is polynomial. The *analytic* theorem is stated over $\mathbb{C}$ and only over $\mathbb{C}$: a field of characteristic zero supplies no absolute value, no square root, and no order *as part of its structure*, so the eigenvector bound of Proposition 8.21, the σ/τ normalisation, and `ENO823Exact` have no meaning over a general $\Bbbk$. In the $\mathbb{C}$ statement the Frobenius–Perron dimensions are a `FPData` on the ring ($\texttt{fpdim} : \mathrm{Fin}\, n \to \mathbb{R}$ with $\texttt{pos} : \forall i,\ 0 < \texttt{fpdim}\, i$), read in $\mathbb{C}$ through the structural inclusion $\mathbb{R} \to \mathbb{C}$, the absolute values are complex moduli, and positivity of a scalar $z \in \mathbb{C}$ is *defined, once*, as "z is real and $z > 0$" (equivalently $\exists\, r \in \mathbb{R}_{>0},\ z = r$). The instances stay over $\mathbb{R}/\mathbb{Q}$ where they are and are joined by $\mathbb{C}$-valued ones where the source demands it; nothing migrates (WP1 Task 1.1, R5).

(S3) **The quadruple-dual isomorphism.** ENO Theorem 2.6 ($g\colon \mathrm{Id} \to (-)^{****}$) rests on the weak Hopf algebra theory of ENO §4–5, which is out of scope. Two honest options are scheduled: take g as a hypothesis and state the theorem conditionally, or formalise Bartlett's string-diagram proof of the same result (Bartlett Theorem 3.28 and Corollary 3.30, §3.6), which avoids weak Hopf algebras entirely. The proposal prefers the second and budgets it; the fallback is the conditional statement, clearly labelled.

(S4) **What is excluded.** The weak Hopf algebra input; the general-multiplicity associator; the classification of rank-three pivotal categories beyond the inequality used as a test; and any claim about ENO Conjecture 2.8 beyond the partial statements listed in WP5.

A theorem proved under (S1)–(S3, first option) is *ENO 8.23 for multiplicity-free data, conditional on g*. The proposal says so on every statement that depends on it, and the deliverable list distinguishes the conditional and unconditional forms.

3 Prior work

3.1 The mathematical sources

The source theorem. The target is Etingof–Nikshych–Ostrik, *On fusion categories*, Annals of Mathematics **162** (2005), 581–642 [ENO]. The relevant material is §2.2 (the quadruple dual),

§3 (the T-matrix and the bound), and §8.4 (pseudo-unitarity and the spherical structure). Two numbering conventions circulate: the published Annals version states the result as Proposition 8.23, while the arXiv version (`math/0203060v2`) states it as Proposition 8.22, one lower throughout §8.4. This proposal follows the Annals numbering and the library cites the number on every use; mis-numbering this proposition has already caused one wrong citation in the project's own plan.

The textbook treatment. EGNO, *Tensor Categories* [EGNO], states the theorem as Proposition 9.5.1 (its notes record that this is ENO 8.23), with pseudo-unitarity defined in Definition 9.4.4 and the global dimension defined through squared norms in §9.4 — the non-circular definition this proposal adopts in Definition 2.7. It is the more readable treatment, and it is the source of the convention that a pivotal structure is a monoidal isomorphism $\mathrm{Id} \to (-)^{**}$ rather than a choice of duals, which matters: the two formulations differ by a twist.

An independent, coordinate-free proof. Bartlett, *Fusion categories via string diagrams* [Bar15], gives a graphical development of exactly the results this project needs, and is the single most useful prior work for the proposed derivation of J:

- Theorem 3.28 and Corollary 3.30: the quadruple dual functor is monoidally isomorphic to the identity (ENO Theorem 2.6);
- Corollary 3.26: the coherence law for the pivotal symbols, in the form $\varepsilon^i_{jk,\alpha}\varepsilon^m_{kl,\beta} = \varepsilon^i_{nl,\gamma}\varepsilon^n_{jk,\delta}$ *whenever* $(F^i_{jkl})^{\gamma n\delta}_{\alpha m\beta} \neq 0$. The nonzero-entry qualification is part of the statement: without it the F-free coherence law cannot be obtained by cancelling an associator entry that may vanish (§4.4);
- Theorem 4.17: an *explicit formula* for the pivotal operators T^i_{jk} — the monoidal coherence of the canonical pivotal endofunctor — as the product of the *apex associator monodromy* and the *pivotal indicators*; this is the closed form the present project is missing for J;
- Theorem 5.4: pivotal structures on a (unitary) fusion category correspond exactly to tuples $(\gamma_i) \subset U(1)$ satisfying $\gamma_j\gamma_k = \gamma_i T^i_{jk}$ on every fusion channel, and sphericality is $\gamma_i = \pm 1$ for all i;
- Corollary 5.5: a pseudo-unitary fusion category admits a canonical spherical structure (the same statement as the target);
- §5.3: the *sphericalization* $\widetilde{\mathcal{C}}$ of $\mathcal{C}$ by the canonical $\mathbb{Z}/2$-action — an *equivariantization*, not a $\mathbb{Z}/2$-graded extension, and the correct description of the cover construction in WP5.

Theorem 5.4 is the cleanest available check on any derived formula for J: the two conventions are reciprocal (the present library's $p_k = J^k_{ij}p_ip_j$ against Bartlett's $\gamma_i = \gamma_j\gamma_k/T^i_{jk}$), and an index-level comparison gives an independent confirmation that does not go through the snake axioms at all. WP3 includes this comparison as an acceptance test.

Applications and the shape of the literature. Ostrik [Ost13] analyses pivotal fusion categories of rank three and derives the inequality

$$\sum_i d_i^2 \;\geq\; \sum_i \bigl|\textstyle\sum_j N^k_{ij} d_j\bigr|^2 / \ldots$$

that pins down the possible dimension functions; it is the place where "pivotal + positive + spherical" is used as a computational tool rather than as a structural hypothesis, and it is the template for the negative tests of WP6. The Haagerup–Izumi programme and the author's own work [Hua26] use the theorem at a different point, to convert coefficient identities (analytic) into a spherical structure (algebraic). Frobenius–Schur indicator theory [NS] is the other major consumer of the same layer: indicators are traces of pivotal structures composed with the duality, and they are generally *cyclotomic* integers — integrality of the higher indicators is a separate arithmetic statement, not a consequence of sphericity, and nothing in this proposal depends on it.

What is *not* in the literature. (a) No proof of ENO Conjecture 2.8 (every fusion category is pivotal); the spherical version is strictly stronger and equally open. The counterexamples to naive expectations are of two kinds: categories with no pivotal structure at all are not known, but categories where the dimension depends on the pivotal structure are *abundant* (the present project has a kernel-checked one, §4). (b) No computation of the double-dual twist J in closed form for general multiplicity-free data; Bartlett's Theorem 4.17 is the closest, and it is stated through graphical monodromy rather than as an F-matrix formula ready for formalisation.

3.2 The formalisation landscape

Mathlib. `CategoryTheory.Monoidal.Rigid.Basic` provides `RigidCategory` (left and right rigid monoidal categories), with the file itself listing "define pivotal categories" as future work. There is no pivotal, spherical, or fusion category in Mathlib at the pinned revision, and no Perron–Frobenius theory for general irreducible non-negative matrices (only `Matrix.IsIrreducible`). The relevant upstream work is in flight:

- **PR #42145**, *the double right dual functor is monoidal* (+233/-4): defines `doubleRightDualFunctor` and its monoidal structure;
- **PR #42150**, *pivotal categories* (+370/-13 in 5 files): defines `PivotalCategory` as a rigid category with a monoidal natural isomorphism between the double dual functor and the identity; currently open and labelled *blocked-by-other-PR* on #42145;
- **PRs #42191** (spherical) and **#42192** (symmetric rigid is spherical), which build on #42150;
- the Perron–Frobenius chain **#39914**–**#39922**, whose declarations `PerronFrobenius.Irreducible`, `pft_irreducible`, `collatzWielandt` and `spectralRadius` supply the spectral theory the fusion-matrix argument needs.

The project's convention is to vendor a byte-compatible copy under `FusionCategory/MathlibPatches/` and to delete it, importing instead, when the pull requests merge. At the time of writing the vendored copies are *placeholders*, not code: they declare nothing that the project uses, `PivotalCategory.lean` is 33 lines and `PerronFrobenius.lean` is 22, and no theorem in the library depends on either. This is honest debt, and WP0/WP5 repay it.

Two claims about the library are kept apart throughout. *Green build*: `lake build` succeeds on the working tree at `d3d320e` (3,032 jobs, warnings only) and `AxiomAudit.lean` reports the permitted axioms on 73 declarations; this was observed. A *reproduced clean build* — a fresh clone, a fresh Mathlib cache, no path override — has not been observed, and is deliverable D2. Neither is evidence that the statements correspond to their sources; that is Gate 3 of Appendix B, and four failures of it are recorded in §4.

Other formal and semi-formal resources. Alphabeta Math's generated library page *Unitary Fusion Spaces, F/R Symbols, and Anyon Models* [AM] has multiplicity-aware fusion trees, the F-symbol as a recoupling matrix, the pentagon in fusion-tree coordinates, gauge covariance, and explicit Fibonacci and Ising carriers; it is pipeline-generated with provenance labels and is not a proof assistant development. TNLean [TN] is a blueprint project on tensor networks whose anyon chapter treats quantum-double models — a different object (microscopic lattice models rather than abstract fusion data). AnyonWiki [AW] supplies F/R/pivotal tables in a fixed gauge for multiplicity-free categories up to rank seven; it is *data*, not code, and the present project's Fibonacci and Ising conventions follow it rather than being derived afresh.

Conclusion. There is no formalisation of pivotal or spherical fusion categories anywhere, and the categorical infrastructure is landing in Mathlib during the period of this proposal. The strategic consequence is fixed: keep the skeletal layer aligned with #42150/#42191 naming, stub the bridge, and delete the patch when it merges. Reimplementing rigid category theory would be wasted

effort; the value added here is the finite-data layer, the examples, the counterexamples, and the theorem.

3.3 The author's prior formalisation

The sibling package `haagerup-odd/` in the same repository is a completed formalisation of the main theorem of [Hua26]: `exists_EGEquations_odd`, the existence of the Evans–Gannon equations for the cyclic Haagerup–Izumi categories at every odd order. It comprises 115 modules and 24,295 lines, with 1,162 theorems and lemmas, no `sorry`, and an axiom audit reporting `[propext, Classical.choice, Quot.sound]`. It is pinned to the same toolchain as this project (Lean `v4.33.1`, Mathlib `0df444a3`).

Its relevance to this proposal is threefold. First, *feasibility*: the proposed library is roughly a fifth of that development in size, and the working method (finite data, exact arithmetic, per-file builds, an axiom audit in CI) is the one already proven at that scale. Second, *conventions*: both packages present fusion data as F-symbols with fixed gauges, and the Haagerup–Izumi coefficients are the natural large-scale instance for WP6. Third, *scope discipline*: that development stops at Chapter 9 of the source manuscript and excludes the categorical reconstruction, exactly as this one excludes the weak Hopf algebra input of ENO §4–5.

4 Preliminary results

This section reports what is already proved, in enough detail to be checked, and marks each claim with its epistemic status:

Status	Meaning
kernel	proved in Lean 4; in `AxiomAudit.lean`; axioms within budget
kernel (witness)	a structure is exhibited and its fields are proved in Lean
script	verified by exhaustive computation in Python (`FusionCategory/docs/*.py`), not formalised
conjecture	stated in `PLAN.md`, no proof

4.1 The formal signature

The library separates the ring from the category. `FusionRing n` carries $N\colon \operatorname{Fin} n \to \operatorname{Fin} n \to \operatorname{Fin} n \to \mathbb{N}$, the involution `dual`, the unit at 0, associativity, and the two duality axioms. `FusionSystem n k` extends it with the categorical data:

Field(s)	Content
`mult_free`	$N_{ij}^k \le 1$
`F`, `F_support`	the associator $F_d^{abc}(e,f)$, vanishing off the channels
`pentagon`	the ten-label pentagon
`F_unit_left`, `F_unit_right`	unit normalisations at $a=1$ and $c=1$
`Finv`, `Finv_support`, `Finv_comp`, `F_comp_inv`	the inverse associator and its two inverse laws
`ev`, `coev`, `ev_ne_zero`, `coev_ne_zero`	the duality scalars
`snake_left`, `snake_right`	the two snake equations
`J`, `J_ne_zero`, `J_unit_left`, `J_unit_right`, `J_dual`, `J_coherent`	the double-dual twist and its four laws

A pivotal structure is then a family $p\colon \operatorname{Fin} n \to \Bbbk^\times$ with $p_0 = 1$ and $p_k = J_{ij}^k p_i p_j$ on every fusion channel; the dimension is $\dim_i = p_i \kappa_i$ with $\kappa_i = \mathrm{ev}_i \mathrm{coev}_i$; sphericity is $\dim_{i^*} = \dim_i$.

The one design decision that deserves comment is `Finv`. The associator at fixed (a,b,c,d) is a square matrix whose rows are the channels of $b \otimes c$ and whose columns are the channels of $a \otimes b$; these are *different* subsets of $\operatorname{Fin} n$, so there is no canonical way to present the inverse as a matrix indexed by $\operatorname{Fin} n$. The project therefore carries α^{-1} as data with its two inverse laws. This is not

pedantry: the right snake equation needs the entry $(\alpha^{-1})^{(0,0)}$, and getting this wrong excluded Fibonacci and Ising from the formalism (§4.3).

4.2 Four defects, and what caught them

	Symptom	Diagnosis	Regression test
D1	every theorem vacuous	unit axioms inconsistent at $(0,b,0,d,b,0)$ for $n \geq 2$	`VecZ2.system`
D2	"pentagon" fails on $\mathrm{Vec}^{\omega}_{\mathbb{Z}/2}$, all ω	eight free labels instead of ten	512-tuple check
D3	target provable with no hypothesis	`p_fusion` fixed $J \equiv 1$	`VecZ2Twisted` witness with $\kappa_g = -1$
D4	Fibonacci, Ising uninhabitable	reciprocal used for matrix inverse in `snake_right`	Fib, Ising

D1–D3 share a signature: a statement that *looks* like its source but quantifies over an empty or over-constrained type. D4 is different — a genuine mathematical error in a normalisation — and it is the one that took three attempts (§4.3). The lesson recorded in the plan is that "the build is green" certifies nothing about the statement; only inhabitation and an independent computation do.

4.3 The duality normalisation

Theorem 4.1 (Snake normalisation; kernel)**.** *Let* $\kappa_i = \mathrm{ev}_i \,\mathrm{coev}_i$. *In a skeletal rigid datum the two snake equations are equivalent to*

$$\kappa_i \; F_i^{i\,i^*\,i}(0,0) = 1 \qquad \textit{and} \qquad \kappa_i \; \left(F_{i^*}^{i^*\,i\,i^*}\right)^{-1}(0,0) = 1,$$

where the second factor is the $(0,0)$*-entry of the* matrix inverse *of the associator block, not the reciprocal of the single entry* $F_{i^*}^{i^*ii^*}(0,0)$.

The two readings agree exactly when the block is 1×1 — true for every pointed category, which is why the earlier "κ/F" correction sufficed for $\mathrm{Vec}^{\omega}_{\mathbb{Z}/3}$ — and differ as soon as a self-dual simple has a 2×2 block. Fibonacci is the minimal case: with the gauge $F_\tau^{\tau\tau\tau} = \left(\begin{smallmatrix} 1/\varphi & 1 \\ 1/\varphi & -1/\varphi \end{smallmatrix}\right)$ the left snake gives $\kappa_\tau = \varphi$, the reciprocal reading demands $\kappa_\tau = 1/\varphi$, and the matrix-inverse reading gives $\varphi \cdot (1/\varphi) = 1$ because the block is an involution. The library proves that the wrong reading fails (`Fibonacci.swapped_Finv_comp_ne_one`), so the correction is itself regression-tested.

Consequences, all kernel-checked:

- `FibonacciSystem : FusionSystem 2` and `IsingSystem : FusionSystem 3`, both over $\mathbb{R}$, with every field proved: pentagon, both unit laws, both inverse laws, both snakes;
- the pivotal structure $p \equiv 1$ is spherical for both, with $\dim = (1, \varphi)$ and $\dim = (1, \sqrt{2}, 1) =$ FPdim;
- $\mathrm{Vec}_{\mathbb{Z}/2}$ still inhabits the structure, so the correction did not buy the examples at the cost of the trivial case;
- the finite checks are done by *exact* arithmetic in $\mathbb{Z}$: each entry is an integer pair (p, q) denoting $(p + q\sqrt{d})/2$ in $\mathbb{Q}(\sqrt{d})$, and `decide` verifies the identities. There is no `norm_num` over the reals and no `native_decide`.

The cost is compile time: the file takes about 100 seconds, so the package build is about three minutes.

Remark 4.2 (Statement of the snake in the general case). The same computation in a non-skeletal category produces two extra structural maps (the duality comparison and the unit constraint). The skeletal form above is the image of the general snake under the equivalence, and the correspondence is the content of WP5's bridge; it is *not* obvious and should not be assumed silently.

4.4 The double-dual twist, and the two identities

Theorem 4.3 (Pivotal dimension is a coboundary; kernel)**.** *Let P be a pivotal structure and define the normalised twist*

$$\tau_{ab}^{c} \;=\; J_{ab}^{c}\cdot\frac{\kappa_c}{\kappa_a\kappa_b}.$$

Then for every fusion channel,

$$\dim_P(c) \;=\; \tau_{ab}^{c}\;\dim_P(a)\,\dim_P(b).$$

Consequently $\dim_P$ *is multiplicative on the fusion ring if and only if*

$$\sum_c N_{ab}^{c}\,J_{ab}^{c}\,\kappa_c \;=\; \kappa_a\kappa_b \qquad \textit{for all } a,b \qquad \textbf{(M)} \tag{6}$$

and ENO 8.23 is equivalent to the per-channel identity

$$J_{ab}^{c} \;=\; \frac{f_c\,\kappa_a\kappa_b}{\kappa_c\,f_a f_b} \qquad \textit{on every channel with } N_{ab}^{c}>0. \qquad \textbf{(K)} \tag{7}$$

The equivalence is two theorems in the library, not one: `eno823_of_twistNorm` proves (K) ⇒ ENO 8.23, and `twistNorm_eq_fpdim_ratio_of_dim_eq` proves the converse. Together with `mult_iff_twist` ((M) ⇔ multiplicativity of $\dim_P$) they make the target a statement about (N, J, κ) alone.

Two consequences deserve emphasis.

(i) **(K) is not invariant under the vertex gauge.** Rescaling $\mathrm{coev}_i \mapsto \lambda_i\mathrm{coev}_i$, $\mathrm{ev}_i \mapsto \mathrm{ev}_i/\lambda_i$ preserves the snakes and fixes κ, but multiplies the defect $\Delta_{ab}^{c} = J_{ab}^{c}\kappa_c f_a f_b/(f_c\kappa_a\kappa_b)$ by $t_a t_b/t_c$ with $t_i = \lambda_i/\lambda_{i^*}$. So (K), hence the target, survives only the duality normalisation $\lambda_{i^*} = \lambda_i$. The repair is to add $\mathrm{ev}_{i^*} = \mathrm{ev}_i$ as an axiom, or to define $\dim_i = p_i\,\mathrm{coev}_i\,\mathrm{ev}_{i^*}$. This is a defect of the signature, not of the mathematics, and WP1 fixes it.

(ii) **On the coevaluation channel (K) is J-free.** On $(i, i^*; 1)$ the axiom $J_{ii^*}^{1} = 1$ applies, so (K) reads

$$\kappa_i\,\kappa_{i^*} = f_i^2 \quad \text{for all } i, \qquad \textbf{(C)} \tag{8}$$

which is exactly sphericity of $p = f/\kappa$. Hence (K) ⇒ (C) is a two-line theorem. The converse is *not* a theorem of the signature: (C) constrains κ alone, while (K) constrains J on every channel, and the pointed datum $\mathrm{Vec}^{\omega}_{(\mathbb{Z}/2)^2}$ with $J \equiv 1$ satisfies the four twist laws and (C) and fails (K) (§4.5). What is true is that (C) is necessary for the target, and that (C) ⇒ (K) holds for the *genuine* J; supplying that implication — which is ENO 8.23 — is what the derivation of J has to deliver. It is not a step that can be taken in the signature as it stands.

4.5 Refutations

Theorem 4.4 (Yang–Lee; kernel)**.** *There is a complete skeletal rigid datum* `YangLee.system` *over* $\mathbb{R}$ *— Fibonacci's fusion ring, associator* $F_{\tau}^{\tau\tau\tau} = \left(\begin{smallmatrix} -\varphi & 1 \\ -\varphi & \varphi \end{smallmatrix}\right)$, $\mathrm{ev} \equiv 1$, $\mathrm{coev}_\tau = -1/\varphi$, *whence* $\kappa = (1, -1/\varphi)$ *— satisfying the pentagon on all* 2^9 *label tuples, both unit laws, both inverse laws and both snakes, such that*

$$\neg\, \textit{ENO823 YangLee.system FibonacciFPData}.$$

Indeed $\kappa_\tau\kappa_{\tau^*} = 1/\varphi^2 \neq \varphi^2 = f_\tau^2$, *and on the channel* $(\tau, \tau; 1)$ *the axiom* $J_{\tau\tau}^{1} = 1$ *forces* $1 = \varphi^4$ *for any pivotal structure with* dim = FPdim.

Three things follow, and they shape the whole programme.

(a) ENO 8.23 is **false** as an implication from the axioms of a rigid fusion datum. The missing hypothesis is pseudo-unitarity, and no condition on J can replace it: the contradiction uses only $J_{\tau\tau}^{1} = 1$, which is an axiom.

(b) The counterexample is *two-simple*. There is no hope that a rank threshold excludes it, and any formalisation that omits positivity will silently prove a false theorem — which is what the first version of this project did.

(c) Yang–Lee is not pseudo-unitary, which is exactly ENO's own remark. The formal statement `YangLee.not_pseudoUnitary` records the numerical failure $\sum_i \kappa_i \kappa_{i^*} = 1 + \varphi^{-2} \neq 1 + \varphi^2$.

Theorem 4.5 (The dimension is not determined by the ring; kernel (witness))**.** $\mathrm{Vec}^{\omega}_{\mathbb{Z}/2}$ *with the non-trivial cocycle ($F = -1$ on the $g \otimes g \otimes g \to g$ block, duality normalisation $\kappa_g = -1$) is a* `FusionSystem 2` *over* $\mathbb{R}$*, carrying two pivotal structures: $p \equiv 1$, with* $\dim g = -1$*, and $p_g = -1$, with* $\dim g = 1$*, which is spherical. The conclusion of ENO 8.23 holds for the second and fails for the first.*

Theorem 4.5 is the counterexample to "the dimension is a function of the fusion ring". It is *not* a statement about the number of pivotal structures: both this datum and the untwisted $\mathrm{Vec}_{\mathbb{Z}/2}$ carry exactly two, and indeed once one pivotal structure exists the set of them is a torsor under the characters of the universal grading group (WP4), so the count cannot distinguish the two. What the witness establishes is the dependence of the *dimension function* on the choice, and hence why pseudo-unitarity is the right hypothesis rather than a technical convenience: it selects, among the pivotal structures, the one with positive dimensions.

Remark 4.6 (A third witness — corrected and kernel-checked by the batch). For $\mathrm{Vec}^{\omega}_{(\mathbb{Z}/2)^2}$ with $\omega = (-1)^{a_1 b_2 c_1}$ one has $\kappa = (1, 1, 1, -1)$, which is not a character of $(\mathbb{Z}/2)^2$. With $J \equiv 1$, (M) fails — kernel-checked, with the refutation of the target under a *satisfied* pseudo-unitarity hypothesis, in `Examples/Z2SqFalseJ.lean`; with $J = \delta(\kappa) = \kappa_a \kappa_b / \kappa_c$ it holds (verified in the census; its kernel check is the batch's one open follow-up). The census itself is `FusionCategory/docs/pointed_twist_check.py`, now exact, reference-checked and self-tested. It is the reason the signature must not add $J^2 = 1$ as an axiom — and the witness for that has been *corrected* by the batch's census repair: the $\omega = \zeta^{a_1 b_2 c_1}$ family on $(\mathbb{Z}/3)^2$ originally quoted here is a **coboundary** (kernel-checked: `Witnesses/Z3sqTwist.lean`, `om\_is\_coboundary`, with the explicit trivialising μ), so its $J^2 \neq 1$ is a statement about the gauge, not the category. The corrected family is $\omega = \mathrm{carry} \times \mathrm{carry} \times \zeta^{a_1 b_2 c_1}$: genuinely twisted (`Witnesses/Z3sqTwisted.lean`, `no\_trivialising\_cochain`), with κ not a character and J^2 a primitive cube root of unity on 36 of the 81 channels — and that category is unitary, so such an axiom would exclude a category to which the theorem applies.

Remark 4.7 (The census, after the batch). The script's census is weaker than it looks, and the proposal does not rely on it. The cyclic families $\mathrm{Vec}^{\omega}_{\mathbb{Z}/n}$ are run with a "carry cocycle" whose exponent is a multiple of n, so the cocycle is identically 1 up to floating-point error; and for cyclic groups in the standard gauge κ is a character, so $J = \delta(\kappa) = 1$ and the census cannot separate the honest twist from the trivial one. The only data sets that test a non-trivial κ are $\mathrm{Vec}^{\omega}_{(\mathbb{Z}/2)^2}$ and $\mathrm{Vec}^{\omega}_{(\mathbb{Z}/3)^2}$, and neither is kernel-checked. Repairing this census was WP3 Task 3.4 — *done* in the batch landing: `FusionCategory/docs/pointed_twist_check.py` is now exact, reference-checked and self-tested, with six genuinely non-trivial families.

4.6 ENO Proposition 8.21

Theorem 4.8 (ENO 8.21; kernel, modulo `DimEigen`)**.** *Let R be a fusion ring with* `FPData` *and let $d\colon I \to \mathbb{R}$ satisfy $\sum_j N_{ij}^k d_j = d_i d_k$ for all i, k. Then $|d_i| \leq f_i$ for all i. If moreover $\sum_i |d_i|^2 = \sum_i f_i^2$ then $|d_i| = f_i$ for all i.*

The proof formalised is ENO's: the weighted norm $\|x\| = \sum_j f_j |x_j|$ is submultiplicative on the eigenvector equation, and `fpdim_char` — the fact that f is a *character*, $f_i f_j = \sum_k N_{ij}^k f_k$ — bounds it by f_i. Two index conventions occur (N_{ij}^k against N_i^{jk}), and the library proves the cyclic symmetry $N_{ij}^k = N_{jk^*}^{i^*}$ (the `cyclic1` statement) needed to pass between them (`cyclic`, `cyclic1`, `cyclic2`); the repo's axioms do not otherwise connect the second and third slots.

The equality case is `eq_abs_of_sum_sq`, and `abs_dim_le_fpdim_of_pseudoUnitary` discharges `DimEigen` from multiplicativity, so ENO's $|V|^2 = \mathrm{FPdim}(V)^2$ is available for any pivotal structure whose dimension is multiplicative.

Remark 4.9 (What 8.21 does not give). It is tempting to think that the T-matrix route bypasses multiplicativity. It does not. ENO's Remark 3.1 says that in the pivotal case the signs ε_i and b_j^{ik} are 1, so $T_j^{ik} = N_j^{ik}$ and the eigenvector equation $T_i d = d_i\, d$ *is* multiplicativity of the dimension. The library proves this identification (`tMatrix_mulVec`, `abs_tMatrix`: the domination $|(T_i)_{kj}| \le N_j^{ik}$ is an equality). The non-trivial b-symbols exist only when no pivotal structure is fixed, and their role is to prove $\dim \mathcal{C} \neq 0$ without one.

4.7 ENO's own route: the square root a and the b-cochain

ENO's proof does not start from a pivotal structure — it produces one — so a signature that begins with p cannot express it. The library therefore also carries ENO's input.

Definition 4.10 (kernel, `SqrtData`)**.** *A* square root of g *is an additive isomorphism* $a\colon \mathrm{Id} \to (-)^{**}$ *with* $a^2 = g$ *monoidal, where* $g\colon \mathrm{Id} \to (-)^{****}$ *is ENO Theorem 2.6. In a skeleton, monoidality of* a^2 *has to be stated against* J^2*, because on labels* $(-)^{**}$ *is the identity: the tensorator of* $(-)^{****} = (-)^{**} \circ (-)^{**}$ *is* J^2*, not* J*.*

Given such an a, ENO set

$$b_c^{ij} \;=\; J_c^{ij}\,\frac{a_i a_j}{a_c}, \qquad T_c^{ij} \;=\; N_c^{ij}\, b_c^{ij}, \qquad d_i \;=\; \mathrm{Tr}(a_i) \;=\; a_i \,\mathrm{coev}_i\, \mathrm{ev}_{i^*}.$$

Theorem 4.11 (kernel)**.** *With these definitions:*

(i) $(b_c^{ij})^2 = 1$ *on every channel, and it follows from* $a^2 = g$ *monoidal, not from* $J^2 = 1$ *(`b_sq`);*

(ii) the trace identity $d_i d_j = \sum_c T_c^{ij} d_c$ *is* equivalent *to*

$$\kappa'_i \kappa'_j = \sum_c N_c^{ij}\, J_c^{ij}\, \kappa'_c, \qquad \kappa'_i = \mathrm{coev}_i\, \mathrm{ev}_{i^*},$$

which is (M) with the duality normalisation κ' *(`trace_iff`);*

(iii) $d_i = a_i \kappa'_i$*, and* $d_i = a_i \kappa_i$ *only when* $\mathrm{ev}_{i^*} = \mathrm{ev}_i$ *(`d_eq_kappa`).*

The content of Theorem 4.11 is an identification, not a discovery: ENO's "by the definition of b" step is (M). But the identification is what dissolves an apparent fork in this project's plan. "Derive J" and "assume multiplicativity" are not two research programmes; they are two faces of one theorem, and the condition J has to satisfy is (M) itself.

Remark 4.12 (What must not be assumed). `Cover.lean`'s `S4Data.g_fusion` reads $g_c = g_i g_j$ on channels. That is the $J = 1$ special case and is wrong in general; it must be restated against J^2. Likewise $J^2 = 1$ and $b^2 = 1$ must not be added as axioms (§4.5).

4.8 The instances, and the audit

Datum	Ring	$\Bbbk$	κ	Role
`VecZ2.system`	$\mathbb{Z}/2$	$\mathbb{Q}$	$(1, 1)$	inhabitant; pentagon on 512 tuples
`VecZ2Twisted.system`	$\mathbb{Z}/2$, $\omega \neq 1$	$\mathbb{Q}$	$(1, -1)$	two pivotal structures
`FibonacciSystem`	Fib	$\mathbb{R}$	$(1, \varphi)$	dim = FPdim, spherical
`IsingSystem`	Ising	$\mathbb{R}$	$(1, \sqrt{2}, 1)$	dim = FPdim, spherical
`k2210`	$K(1,0,1)$	$\mathbb{R}$	—	`FPData` certificate; FPdim $= (1,2,1)$, $D^2 = 6$
`YangLee.system`	Fib	$\mathbb{R}$	$(1, -1/\varphi)$	refutes ENO823 without hypotheses

`AxiomAudit.lean` runs `#print axioms` on 73 declarations covering every result quoted above and reports `[propext, Classical.choice, Quot.sound]` throughout. The package builds in 3,032 jobs with warnings only at the recorded commit. There is no `sorry` in the library (the single occurrence of the string is in a comment recording the axiom budget). Two limits of this evidence: the build is a green build of a working tree, not a reproduced clean build (§3), and an axiom audit certifies the *use of axioms*, not the *correspondence of statements with their sources* — the latter is Gate 3 of Appendix B, and it is where the defects of §4 were found.

`ENO823` itself is a `def`, not a `theorem`: it records the open target. The conditional theorem `eno823_of_pseudoUnitary` takes positivity and multiplicativity of a chosen pivotal dimension as hypotheses (`IsPseudoUnitaryPivotal`) together with `PFUnique`, and concludes the theorem. Both hypotheses are necessary: positivity fails on Yang–Lee, and multiplicativity fails on $\mathrm{Vec}^{\omega}_{(\mathbb{Z}/2)^2}$ with $J \equiv 1$.

4.9 The exact gap

Stated as a list, the gap between the present library and the target is:

(A) **The twist is data, not derived.** J is constrained by four laws; the closed form carried in `PLAN.md` (the Ω transcription) is retracted there, and the census that was meant to support it is inconclusive (§4.5, Remark 2.5). So no correct derivation exists yet. Consequences: `p_mon` mentions an undetermined object, and (M) does not follow from the four twist laws as such — a datum may carry an arbitrary coherent J, and on $\mathrm{Vec}^{\omega}_{(\mathbb{Z}/2)^2}$ with $J \equiv 1$ it fails — so (M) has to come from the derivation of J, not from the axioms.

(B) **The analytic step is absent, and its statement needs care.** Positivity is not formalised, and neither is ENO's two-stage normalisation: a *tensor* automorphism $\sigma = d^2/|d|^2$ of the identity, followed by a *natural* automorphism $\tau = d/|d|$ which is not monoidal. Neither is sign-valued a priori (WP4).

(C) **The signature is narrower than the mathematics.** Real rather than complex scalars; multiplicity-freeness assumed; the triangle axiom not stated (it is derivable, WP1); $\mathrm{ev}_{i^*} = \mathrm{ev}_i$ not assumed, so (K) is gauge-fragile. The label question is *not* a restriction, by ENO Proposition 2.1 (§2.3).

(D) **The cover and grading layer is scaffolding.** `coverGlobalDim_double` is a reindexing identity on $\sum d^2$; `coverCanonicalPivotal` returns the input scalar with no `PivotalStructure`; the three former axioms of `Grading.lean` are gone and their content is a theorem plus two `def`s recording open targets.

(E) **No bridge to Mathlib, and no reconstruction.** `MathlibPatches/` contains placeholders, and `FPData` is supplied by hand. The skeletal datum has not been shown to be a fusion category, so every categorical statement about it — including "the categorical dimension" — is presently a statement about scalars.

These five items, together with the reconstruction that gives them meaning (WP2), are the work packages of §5.

5 Research programme

5.1 Aims, and the logical geography of the target

The programme has three aims, corresponding to Q1–Q3, and one exploratory strand for Q4 that is deliberately kept off the critical path.

A1 (Signature and scope).
A corrected signature and an *exact* target: the $\mathbb{C}$ scalar contract for the analytic statements (S2), the triangle law derived rather than assumed, the coherence law with the nonzero-entry

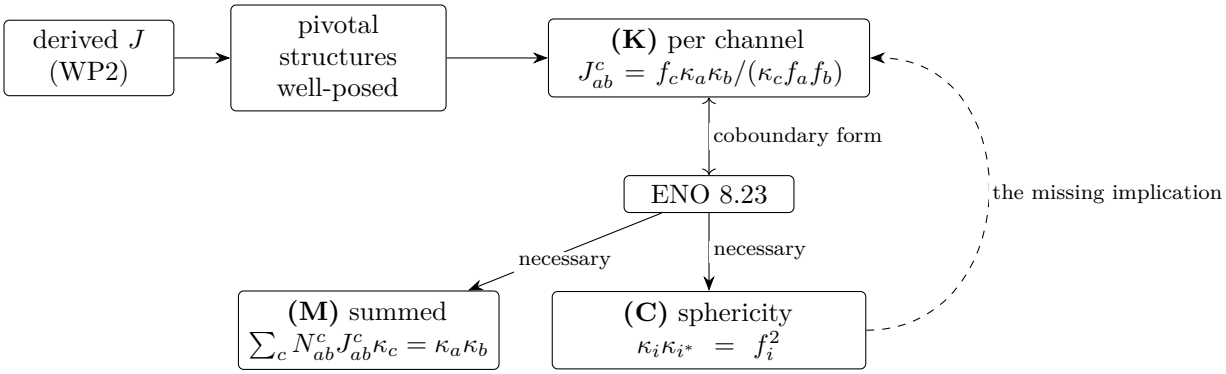


Figure 1: The logical geography, corrected. (K) is the coboundary form of the target and is equivalent to it in the signature. (C) is *necessary* but not sufficient: it constrains κ alone, while (K) constrains J on every channel, and the pointed datum $\mathrm{Vec}^{\omega}_{(\mathbb{Z}/2)^2}$ with $J \equiv 1$ satisfies (C) and fails (K) (§4.5). The same holds for (M), which is weaker still and is satisfied by Yang–Lee. The dashed arrow (C)⇒(K) is *not* a theorem of the signature; it is part of the categorical argument that the reconstruction (WP2) and the derived twist (WP3) have to supply. The derivation of J is logically prior: it makes the other boxes statements about determined objects.

qualification, a gauge-invariant duality normalisation, and a statement of ENO 8.23 whose hypotheses and conclusion match the source line by line. A1 also includes the reconstruction: the skeletal datum must be shown to *be* a fusion category before any categorical word in the statement is meaningful.

A2 (Twist).
The double-dual twist J derived from $(F, \mathrm{ev}, \mathrm{coev})$ — not transcribed from a formula that `PLAN.md` has retracted — with its four laws and its qualified coherence proved, so that "p is pivotal" becomes a well-posed predicate.

A3 (Theorem).
ENO 8.23 for multiplicity-free data: existence, sphericity, positivity, equality with FPdim, and uniqueness, conditional on g or with g proved by Bartlett's graphical argument — algebraic layer over a characteristic-zero field, analytic statement over $\mathbb{C}$ (S2 of §2.6).

A4 (Frontier, exploratory).
The pivotal cover as an equivariantization and the torsor of pivotal structures: statements about *differences* of pivotal structures, which is what the theory actually supplies. No existence criterion is claimed (§5.3).

The logical relations between the statements are worth recording once, because the project's own plan confused them for several weeks. Figure 1 is the map, corrected: the implication that is *missing* is the one that matters.

Three readings of the figure matter for planning. First, **no box in it is the theorem except (K)**, and (K) is a statement about a J that has not yet been derived; the diagram is therefore a map of obligations, not of results. Second, **(C) and (M) are necessary and not sufficient**: each is satisfied by data on which the target fails, so neither can be used as a lemma toward the target in the signature as it stands. Third, **the derivation of J proves nothing about the target by itself**; it removes an undefined object from the statement. This is why A2 and A3 are separate work packages and why neither can be skipped.

5.2 The first delegation batch, and the source of truth

The programme begins with one reviewable batch of infrastructure and statement-settling work, and nothing that depends on it starts until the batch has passed review. The batch:

(B1) **Reproduce the clean build** (WP0 Task 0.2): fresh clone, fresh cache, no path override — the "green build" of §3 becomes a reproducible one, or the reproducibility debt is diagnosed while it is still cheap.

(B2) **Implement the enforcing axiom audit** (WP0 Task 0.2, Gate 4 of Appendix B): wire the checked-in gate (`FusionCategory/tools/audit-gate.sh`, landed with its manifest and twelve-fixture self-test) into CI, and generate the audit list mechanically, so that coverage is not maintained by hand — done in the batch landing: `tools/gen-audit.sh` enumerates every exported declaration from the elaborated environment (1,058 at this revision), `.github/workflows/fusion-category-ci.yml` chains the clean build, the completeness build, the gate and the self-test, and the `set\_option\ linter.auxLemma\ false` line in the generated file keeps the `match\_1` auxiliaries' linter notes out of the gate's input. The `\#print axioms`-plus-`grep` report is withdrawn.

(B3) **Repair the cocycle census** (WP3 Task 3.4, pulled forward): genuine cocycles in place of the identically-trivial carry cocycle, and the two script-level witnesses kernel-checked.

(B4) **Settle the corrected signature and the theorem statement** (the *statements* of WP1 Tasks 1.1–1.5): record the choices — the scalar contract, which coherence form the signature restates, the gauge normalisation, and the exact target with its two qualifications from Task 1.5 (the conditional equivalence with the bare form, and the `J_match`/derived-twist form of the antecedent; the `derivedJ` *definitions* remain WP2 Task 2.3, held by the rule below). Proofs and the six instance re-verifications stay in WP1/M2.

Source of truth. This document, at this revision, is the source of truth for the programme: work packages, theorem contracts, gates, and scope. The implementation-level `PLAN.md` predates it, and this revision supersedes several of its instructions — among them the request for a nontrivial double-dual label permutation (withdrawn: ENO Proposition 2.1 fixes isomorphism classes, Remark 2.6) and the description of positivity as coming from a Hermitian structure (removed: positivity is part of what ENO's σ/τ argument produces, §1.5). Where the two documents conflict, this proposal governs; `PLAN.md` is brought into conformance in the same revision, its corrections are visible in the commit history, and its original step list is retained under a superseded banner rather than rewritten. Agents delegated work on this programme are given this paragraph verbatim.

Holding rule. Dependent theorem work — the reconstruction proofs of WP2, the derivation of WP3 (Tasks 3.1–3.3), and the normalisation of WP4 — does not begin until the batch above has passed review. The rule is deliberate: every defect of §4.2 was a statement settled while proof work was already under way, and the reviews of this proposal found the same pattern in its own instructions.

5.3 Work packages

The order below is the order of the schedule and of the risk: nothing in WP3–WP4 is well-posed until WP1 has fixed the statement and WP2 has shown that the objects in it are categorical.

WP0 — Foundations and reproducibility (months 0–4)

Objective. Replace hand-supplied inputs with proved or explicitly labelled ones, and make the build reproducible from a clean checkout.

Tasks.

0.1 Vendor the Perron–Frobenius chain (#39914–#39922) and define FPdim as the Perron root of the fusion matrix; prove Theorem 2.3 and delete `FPData` where it is not needed. Where uniqueness is genuinely required, keep the weaker `PFUnique` already in `PFDim.lean`; otherwise

`FPData` stays an explicit hypothesis with a per-example certificate, as the FP-dimension subsection of §2 records.

0.2 Clean-tree CI: fresh clone, `lake exe cache get`, `lake build`, and the *enforcing* audit gate of Gate 4 (Appendix B) — the checked-in script `FusionCategory/tools/audit-gate.sh` with its manifest and self-test, not the `\#print axioms` report — as separate gates, each of which fails the build. A from-clean build of the committed tree is now *reproduced* (batch B1: fresh clone from GitHub, a genuinely fresh Mathlib cache — 8690 artifacts into a cache directory that did not exist before, no path override; content-reproducible, not byte-reproducible, so the gates judge the audit's content and never artifact hashes). This is also what turned the "green build" of §3 into a reproducible one; it is batch item B1 (§5.2).

0.3 A per-file build harness with a timeout, so that a 100-second example file does not block work on the core.

0.4 `BLUEPRINT.md` with the module dependency graph (Figure 2), the statement-to-source table of Appendix B (Gate 3), and the scope labels (S1)–(S4) attached to every theorem that depends on them.

Deliverable. D1, D2. **Success criterion.** A machine with the toolchain and the Mathlib cache available builds the library and passes the audit from a clean clone. (The batch's reproduction found the honest form of this criterion: the Mathlib cache server is a *hard* network dependency of a fresh machine — a network-free build requires the cache to be pre-seeded — and the artifacts are content-reproducible, not byte-reproducible; the kit gates on the audit's content, never on artifact hashes.)

WP1 — The corrected signature and the exact target (months 1–8)

Objective. A1: make the finite-data signature express the mathematics, and make the formal statement of the theorem say exactly what ENO and EGNO say.

Tasks.

1.1 **The scalar contract.** Keep the algebraic layer — `FusionSystem`, `PivotalStructure`, the twist reduction, the census — over a field $\Bbbk$ of characteristic zero, where every obligation is polynomial. State the *analytic* target over $\mathbb{C}$ and only over $\mathbb{C}$. Concretely: define, once, the positivity predicate `Pos` on $\mathbb{C}$ — `Pos` $z := \exists r : \mathbb{R},\ 0 < r \wedge z = r$, marked `noncomputable` — and state the $\mathbb{C}$ target as *new* declarations alongside the retained $\mathbb{R}$ originals: `IsPseudoUnitaryC` and `ENO823ExactC` (the rendering of Appendix C with `Pos (dim P i)` in place of `0 < dim P i`), and the $\mathbb{C}$ forms of `pivotal_ext`, `IsPseudoUnitaryPivotal` and `eno823_of_pseudoUnitary` (their proofs are field-generic, but the $\mathbb{R}$ statements cannot simply be generalised: the order, the modulus and the positivity conjunct fail). The bare `ENO823` is already k-polymorphic with `[Algebra R k]` and elaborates at $\mathbb{C}$; the $\mathbb{R}$ originals are *retained* — three of the six data (Fibonacci, Ising, Yang–Lee) instantiate them directly; the two $\mathrm{Vec}_{\mathbb{Z}/2}$ data are $\mathbb{Q}$-valued and $K(1,0,1)$ is a ring, so those three need the $\mathbb{Q} \to \mathbb{R}$ (or $\mathbb{Q} \to \mathbb{C}$) transport, which is part of this task. The port of the analytic chain is an explicit work item: the 8.21 bound and its equality case (`abs_le_fpdim_of_eigenvector`, `eq_abs_of_sum_sq`, at present $\mathbb{R}$-valued) restated with the complex modulus, and the `PFUnique` step applied once `Pos (dim P i)` is turned into a real positive witness. The instances are not migrated: the $\mathbb{R}/\mathbb{Q}$ ones stay, and $\mathbb{C}$-valued instances are added where the source demands them (§2.6, S2; R5). (Two earlier drafts of this task were wrong about the tree: `ENO823Exact` is at present $\mathbb{R}$-only — not "a statement about a `FusionSystem` over $\mathbb{C}$" — and the bare form does typecheck over a general field.)

1.2 **The triangle law, derived.** The middle-unit entry $t = F_d^{a\,1\,c}(a,c)$ is *not* independent of the present axioms: the pentagon at $(a,1,1,c)$ gives $t = t^2$; the block $F_d^{a\,1\,c}$ is 1×1 (the unit

laws force the channels $e = a$ and $f = c$), so the inverse law `Finv_comp` gives $\texttt{Finv} \cdot t = 1$ and hence $t \neq 0$; therefore $t = 1$. Prove it as a lemma. (An earlier draft said "invertibility of the associator gives $t = 1$"; an invertible matrix may have vanishing entries, and the nonvanishing here is the 1×1 block plus the inverse law. The earlier plan to add the axiom was wrong about its independence.)

1.3 **Coherence with the nonzero-entry qualification.** The scalar coherence law may only cancel an associator entry that is nonzero; Bartlett's Corollary 3.26 states the pivotal-symbol coherence exactly that way. Neither support axiom supplies the qualification, and an earlier draft of this task wrongly said they did: `F_support` forces zeros *off* the admissible channels and says nothing about the entries on them, and `J_ne_zero` is a statement about J, not about F. So the corrected `J_coherent` takes one of two forms, both acceptable: (i) the qualified law, carrying an explicit side condition $h_F : F_d^{abc}(e, f) \neq 0$ at the entry to be cancelled, from which the present F-free law follows on that channel; or (ii) the law with the F-factors retained,

$$J_{ab}^{e}\, J_{ec}^{d}\, F_{d}^{abc}(e, f) \;=\; J_{bc}^{f}\, J_{af}^{d}\, F_{d}^{abc}(e, f),$$

from which nothing is cancelled and the F-free law is a corollary under h_F. The restatement is a *weakening of the field* — `J_coherent` as it stands is replaced, not kept alongside — and it is conservative: no theorem in the library consumes the field, and the five instances close either form by `simp`. The citation is for the *shape* of the law: Bartlett's Corollary 3.26 is stated for pivotal symbols (his T-symbols, with $T^2 = \mathrm{id}$), not for the unnormalised J, which must *not* satisfy $J^2 = 1$; transferring the law to J is WP3's work. The choice of form is recorded in the blueprint, and WP3 proves whichever form is chosen for the derived twist. No proof in the programme cancels an F-entry without an explicit nonzero hypothesis, and no proof derives nonvanishing from `F_support` or `J_ne_zero`.

1.4 **Gauge-invariant duality normalisation.** Either add $\mathrm{ev}_{i^*} = \mathrm{ev}_i$ as an axiom, or define $\dim_i = p_i \,\mathrm{coev}_i \,\mathrm{ev}_{i^*}$ as ENO do. The first makes (K) gauge-invariant; the second matches the source but changes every instance proof. The proposal is to do the first and prove the two normalisations agree in the self-dual case.

1.5 **The exact target: its hypothesis, and which forms are equivalent.** The library's `ENO823Exact` is

$$\texttt{IsPseudoUnitary}\ S \to \exists!\, P,\ P.\mathrm{IsSpherical} \wedge (\forall i,\ \dim_P i > 0) \wedge (\forall i,\ \dim_P i = f_i),$$

which does carry existence. Every statement *about* it must reproduce the existential conjunction verbatim; the earlier rendering of the target in this proposal split it into a spherical-structure field plus separate positivity and uniqueness fields — asserted only conditionally — which is strictly weaker; it is withdrawn and is *not* a proof target (Appendix C). Two qualifications on the statement itself, both fixed in this revision:

(a) **The equivalence with the bare form is conditional.** The bare form `ENO823` ($\exists P$, spherical, $\dim_P = f$) has no pseudo-unitarity antecedent, so `ENO823Exact` does *not* imply it unconditionally: on Yang–Lee the exact form holds vacuously (its antecedent is refuted by `YangLee.not_pseudoUnitary`) while the bare form is false (`YangLee.not_eno823`). What is to be proved is the equivalence *under* the hypothesis: given `IsPseudoUnitary`, exact ⇒ bare is `eno823_of_exact`, and bare ⇒ exact is unconditional, by `pivotal_ext` (a pivotal structure is determined by its dimension function) together with the positivity of the FP dimensions ($\dim_P i = f_i > 0$ supplies the positivity conjunct). State the conditional form `eno823_of_pseudoUnitary` separately; it assumes a pivotal structure rather than producing one and is strictly weaker than both.

(b) **The target is stated for the derived twist.** As displayed, the statement quantifies over the *carried* J, and as such it is *false*: the pointed $\mathrm{Vec}^{\omega}_{(\mathbb{Z}/2)^2}$ datum with $J \equiv 1$ is

pseudo-unitary ($\sum_i \kappa_i\kappa_{i^*} = \sum_i f_i^2 = 4$) and admits no pivotal structure with dim = FPdim — with $J \equiv 1$, `p_mon` forces p to be a character, and $1/\kappa$ is not one. The exact target is therefore stated either in the corrected signature, the carried J field replaced by `derivedJ` (Task 2.3), or with the compatibility hypothesis `J_match` in the antecedent, on the model of (a). Every statement of the target in this proposal (Task 4.6, D6, M6, Appendix C) carries this qualification.

1.6 **The label question, recorded as closed.** Double duality fixes isomorphism classes (ENO Proposition 2.1), so there is no label permutation to add. The plan to add one was based on the error corrected in Remark 2.6; the blueprint records the retraction and no work item remains.

Deliverables. D3. **Success criterion.** All six existing instances re-verified under the corrected signature, with the same axiom budget; the triangle law and the F-free coherence law under h_F proved as lemmas, the coherence axiom itself recorded in the form chosen in the blueprint and supplied by the instances; a target statement that matches Theorem 2.8 field by field, stated over $\mathbb{C}$ under the scalar contract of Task 1.1 and with the derived twist (or `J_match`) in its antecedent (Task 1.5).

WP2 — Reconstruction and the trace correspondence (months 3–10)

Objective. Make the words in the statement mean what they say. Until the skeletal datum is a category, "the categorical dimension" is a scalar defined by a formula, and the theorem is a statement about a formal system.

Tasks.

2.1 **A small reconstruction.** For multiplicity-free data, define the $\Bbbk$-linear category whose objects are formal finite direct sums of simples and whose morphisms are matrices indexed by channels; prove it is monoidal, rigid, semisimple, and that $\mathrm{End}(\mathbf{1}) = \Bbbk$ — and package the four properties as a named predicate (`IsFusionCategory`; no such predicate exists in the library today, so WP2's success criterion and D4 have nothing to check until it is named). The associativity and unit constraints are the F-data and the unit laws; the snakes are the snake axioms. This is the piece of the programme that converts a formal calculation into a theorem about fusion categories, and it is scheduled before the twist because the twist is a morphism of the reconstructed category.

2.2 **The trace correspondence.** Prove that the categorical trace of $p_i \colon X_i \to X_i^{**}$ is $p_i\kappa_i$, that $\mathrm{Tr}(a_i) = a_i \,\mathrm{coev}_i \,\mathrm{ev}_{i^*}$ is ENO's d_i, and that $\kappa_i\kappa_{i^*} = |X_i|^2$ in the sense of Definition 2.7. This is small, checkable, and closes the gap between the library's scalars and the source's dimensions.

2.3 **The twist as a morphism: derived J, and its relation to the carried field.** From the reconstructed category, *define* `derivedJ` as the (k,k)-entry, in the channel basis, of the monoidal structure of $(-)^{**}$ — a specific morphism $X_i^{**} \otimes X_j^{**} \to (X_i \otimes X_j)^{**}$, a matrix indexed by channels (the diagonal reading is forced once a pivotal structure exists, but `derivedJ` is defined before any is known) — so that WP3 derives a morphism and not a table. Do *not* identify `derivedJ` with the datum's carried field $S.J$: that field is free data constrained only by the four twist laws, and the pointed false-J datum ($\mathrm{Vec}^{\omega}_{(\mathbb{Z}/2)^2}$ with $J \equiv 1$, §4.5) satisfies every law the signature imposes while its honest tensorator is $\delta(\kappa)$. The relation between the two is therefore an explicit *compatibility hypothesis*, `J_match`, stating that $S.J$ and `derivedJ` agree on every channel, discharged per instance by WP3 (Task 3.3), or else the corrected signature of WP1 replaces the carried field by the derived definition once WP3 has proved the four laws for it. Reconstruction must not ask for the identification outright: that instruction is contradicted by the false-J example, and an earlier draft of this task gave it.

Deliverables. D4. **Success criterion.** The reconstructed category is a `FusionCategory` in the sense of the library's own definition, and every scalar named in the target is the corresponding trace; `derivedJ` is defined independently of the carried field, and the relation between the two is the explicit hypothesis `J_match` or a field replacement — never an identification by fiat.

WP3 — Deriving the double-dual twist (months 4–14)

Objective. A2: J as a function of $(F, \mathrm{ev}, \mathrm{coev})$, derived and not transcribed.

The retraction this package has to respect. `PLAN.md` records that the displayed formula

$$\Omega(a,b;c) = F_0^{b^*a^*c}(c^*,b)\cdot(M_b)^{-1}(0,c)$$

is *defective as written*: read literally it returns $+1$ on $(\mathbb{Z}/2)^2$ with $\omega=(-1)^{a_1b_2c_1}$, where the honest twist is $\delta(\kappa)$, i.e. -1 on six channels. The construction of the comparison is right and the transcription is not. `FusionCategory/docs/honest\_twist\_check.py` evaluates the twist from its defining property and never evaluates the displayed Ω. No milestone below depends on the displayed formula.

Tasks.

3.1 **The comparison map.** Construct $\psi_{a,b}\colon (a\otimes b)^*\to b^*\otimes a^*$ as the unique map intertwining the evaluations, and prove it is an isomorphism.

3.2 **The ratio formula.** Prove

$$J^c_{ab} \;=\; \frac{\psi_{a,b}|_c}{\psi_{b^*,a^*}|_{c^*}},$$

and *then* derive a closed form in terms of F; if the closed form differs from the retracted Ω, the retraction is the correct record and the new formula replaces it in `PLAN.md` and the blueprint. Here `derivedJ` is the tensorator composite of Task 2.3 (its (k,k)-entry in the channel basis), so the ratio is a *theorem*, and its statement includes the nonvanishing of both components on every supported channel — available because ψ is an isomorphism and the channels are one-dimensional under `mult_free`, not by cancellation.

3.3 **The four laws and the qualified coherence** of WP1 Task 1.3, proved for the derived twist `derivedJ` of WP2 Task 2.3. This is the step that turns "J is data" into "J is computed", and it is independent of pseudo-unitarity. For instances carrying their own J field, the compatibility hypothesis `J_match` of Task 2.3 is discharged here; alternatively the corrected signature of WP1 replaces the carried field by `derivedJ`, in which case there is nothing to match.

3.4 **Repair the census.** The cyclic families in `docs/pointed\_twist\_check.py` use a "carry cocycle" whose exponent is a multiple of n, so it is identically 1, and for cyclic groups κ is a character, so those cases cannot separate the honest twist from the trivial one (§4.5). Replace them with genuine cocycles: $(\mathbb{Z}/2)^2$, $(\mathbb{Z}/3)^2$, a non-trivial 3-cocycle on $\mathbb{Z}/3$, and a non-abelian group; kernel-check the two witnesses that are currently script-level.

3.5 **Independent confirmation.** Compare index by index with Bartlett's Theorem 4.17 and Corollary 3.26, and with the corrected census. A mismatch is a defect in the derivation, not in the check.

3.6 **(M) as an acceptance test.** For the genuine J, (M) should be a theorem — the dimension of a pivotal category is multiplicative. A failure of (M) after the derivation means the derivation is wrong; it is not evidence that (M) is an extra hypothesis.

Deliverables. D5. **Success criterion.** No hypothesis on `derivedJ` beyond the four laws — they are theorems about the derived object; a carried J, where one remains, is linked to it

by `J_match` proved per instance; the census contains at least four genuinely non-trivial cocycles and is kernel-checked. **Explicit non-goal:** WP3 does not prove (K). The target is the strictly stronger per-channel identity, which fails for Yang–Lee and is WP4's content.

Status at closure (swarm `smwa3l4s5`). WP3 is closed at its honest boundary. Every data set in the tree whose carried twist is the honest one carries `J_match` at the derived twist: Fibonacci and Ising (round 1), Yang–Lee (`DerivedJYangLee.lean`: the closed form `derivedJGen = 1` on every supported channel, the nonvanishing in both the scalar and the frozen `OmegaNe` form, the fourth law through `J_match`, the fixpoint datum, and the *settledJ* sharpness $\varphi^{-4} \neq 1$ at $(\tau, \tau; \mathbf{1})$), the pointed systems (`JMatchPointed.lean`: `VecZ2`, `VecZ2Twisted`, `Z3Omega`, `Z2SqH`, and the $\mathbb{C}$ transports). Task 3.4's census is kernel-checked with genuinely non-trivial cocycles ($(\mathbb{Z}/3)^2$ twisted and D_8, `Witnesses/`); Task 3.5's Bartlett Theorem 4.17 comparison is certified channel by channel in the kernel (`DerivedJBartlett.lean`); the double-dual tensorator is an isomorphism (`DoubleDualTensoratorIso.lean`, unconditional on pointed data and instantiated at `Z2SqH`); and the admissible twists of a fixed datum form a group, with `J_match` the torsor statement (`DerivedJTwistGroup.lean`). The four laws hold for the derived object as conditional theorems (the coboundary class, `PairCocycle`, `SingleDualCoherent`, the two-obligation assembly), and the nonvanishing layer is characterised exactly as two inverse-associator entries. The *unconditional* general fourth law is equivalent to ENO Conjecture 2.8 (open) and is recorded as out of reach: no refutation below rank 8 exists in the literature, every family tested empirically is pivotal, and a refutation would need rank ≥ 10, characteristic $\neq 0$, and a non-pointed multiplicity-free associator with a singular inverse-block minor. This boundary — not a defect in the derivation — is what separates the conditional theorems from the unconditional law; the per-instance coverage above is what the tree proves.

WP4 — ENO's normalisation and the theorem (months 8–20)

Objective. A3: ENO 8.23, with the normalisation argument stated as ENO state it.

Tasks.

4.1 **Construct a.** An additive isomorphism $a\colon \mathrm{Id} \to (-)^{**}$ with $a^2 = g$ monoidal. `SqrtData` currently assumes a; this task constructs it from g (Theorem 2.10), which is either a hypothesis (S3, first option) or a theorem of WP4 Task 4.2.

4.2 **The quadruple dual.** Either keep g as an explicit hypothesis, or formalise Bartlett's Theorem 3.28 and Corollary 3.30: the quadruple-dual functor is monoidally isomorphic to the identity, proved by string diagrams (§3.6 of [Bar15]) and avoiding the weak Hopf algebra theory of ENO §4–5. This is the single largest prerequisite in the programme and is budgeted as such; if it slips, the theorem is stated conditionally.

4.3 **The bound.** From the trace identity $d_i d_j = \sum_c T_c^{ij} d_c$ with $T_c^{ij} = N_c^{ij} b_c^{ij}$ and $|b| = 1$, derive $|d_i| \leq f_i$ by ENO's weighted norm. The library has the weighted-norm argument (`abs_le_fpdim_of_eigenvector`); what is missing is the passage from the T-matrix to the N-matrix, i.e. the triangle inequality $|\sum_j T_j^{ik} d_j| \leq \sum_j N_j^{ik} |d_j|$. (Both 8.21 statements are $\mathbb{R}$-valued in the library at present; their $\mathbb{C}$ restatements with the complex modulus are part of Task 1.1's port.)

4.4 **The sum identity.** Prove $\sum_i |d_i|^2 = \dim \mathcal{C}$ for $d_i = a_i \kappa_i'$, which under pseudo-unitarity gives $\sum_i |d_i|^2 = \sum_i f_i^2$; the equality case of Theorem 4.8 then yields $|d_i| = f_i$.

4.5 **The two normalisations, correctly typed.** This is the step the earlier draft of this proposal misstated, and it is worth stating exactly.

(a) The chain $f_i f_j = |d_i d_j| \leq \sum_c N_{ij}^c f_c = f_i f_j$ is an equality. Equality in the triangle bound gives $b_{ij}^c d_c = \varphi_i \varphi_j f_c$ on every channel (with $\varphi_i = d_i / f_i$, $|\varphi_i| = 1$); since moreover $(b_{ij}^c)^2 = 1$ (Task 4.1, `b_sq`: it follows from $a^2 = g$ monoidal, not from the chain),

$T^c_{ij} = b^c_{ij} N^c_{ij} = \pm N^c_{ij}$ and $d_i d_j / d_c$ is real whenever $N^c_{ij} > 0$. Consequently $\sigma_i = d_i^2/|d_i|^2$ defines a *tensor automorphism of the identity functor*. It is not sign-valued a priori — it is a phase — and its monoidality is *equivalent* to a statement about *squares*: on every channel,

$$(d_i d_j/d_c)^2 \;=\; (f_i f_j/f_c)^2 \;>\; 0,$$

since $\sigma_i\sigma_j/\sigma_c = (b^c_{ij})^2 = 1$; so only the squares of the ratios are pinned down and positive at this stage, the ratios themselves being real of fixed modulus and undetermined sign. (Two earlier drafts mis-stated this step: one said the ratios themselves were positive — which presupposes the sign-fixing — and one derived $T^c_{ij} = \pm N^c_{ij}$ from the chain alone, which needs `b_sq`.)

(b) **The normalisation, as one rescaling.** Rescale $a_i \mapsto a'_i = a_i\, f_i/d_i$ (the character $\chi_i = f_i/d_i = \overline{\varphi_i}$; note $\chi^2 = \sigma^{-1}$, so this is ENO's "twist by σ" up to the inverse convention, ENO's $\tau_i = d_i/|d_i|$ naming the residual sign freedom in the choice of χ). Then, directly: $d'_i = a'_i\kappa'_i = f_i > 0$; the defect becomes $b'^c_{ij} = b^c_{ij}\,\chi_i\chi_j/\chi_c = (\varphi_i\varphi_j/\varphi_c)\cdot\varphi_c/(\varphi_i\varphi_j) = 1$ on every channel; and $a'_0 = 1$ (`a_unit`, `kappa_unit`, $f_0 = 1$), so `toPivotal` applies: a' is a pivotal structure with $\dim = f > 0$, spherical because $f_{i^*} = f_i$ (`FPData.dual_eq`). ENO's two-step σ-then-τ description is this single rescaling compressed; an earlier draft of this task reproduced it with "twist g by σ" left undefined — not executable — and it is replaced here by the rescaling.

The point to keep: multiplying a by a monoidal character leaves the defect $b^{ij}_c = J^{ij}_c a_i a_j / a_c$ *unchanged*, so the σ-step cannot remove a defect; it normalises the phase. The τ-step is the one that acts on a, and it is not monoidal. A draft that made both steps "tensor autoequivalences acting by signs" was wrong on both counts. (For a concrete illustration: in $\mathrm{Vec}_{\mathbb{Z}/3}$ the tuple $a = (1, -1, 1)$ satisfies $a^2 = 1 = g$ but is not a character, so rescaling a square root by arbitrary scalars changes the defect.)

4.6 **Assembly and uniqueness.** `ENO823ExactC` (the statement Task 1.1 freezes; the proof's working name is `eno823_exact`): existence, positivity, equality with FPdim, uniqueness via `pivotal_ext` and `PFUnique` — stated as the exact target of Task 1.5: over $\mathbb{C}$ under the scalar contract, with the derived twist (or `J_match`) in the antecedent.

Deliverables. D6. **Success criterion.** `ENO823Exact` proved for multiplicity-free data over $\mathbb{C}$ under the scalar contract of WP1 Task 1.1, with the hypothesis on g either discharged (Bartlett route) or explicitly labelled; `YangLee.not\_pseudoUnitary` as the sharpness witness.

WP5 — Cover, grading, and the frontier (months 14–22, off the critical path)

Objective. A4: the part of the theory that does not need pseudo-unitarity, stated precisely enough to be useful and no more.

Tasks.

5.1 **The cover as an equivariantization.** Bartlett §5.3: the sphericalization $\widetilde{\mathcal{C}}$ is the equivariantization of $\mathcal{C}$ with respect to the canonical $\mathbb{Z}/2$-action, *not* a $\mathbb{Z}/2$-graded extension of $\mathcal{C}$. Prove it is a fusion category, identify the forgetful functor, and prove $\dim\widetilde{\mathcal{C}} = 2\dim\mathcal{C}$. The present `Cover.lean` is scaffolding: `coverGlobalDim\_double` is a reindexing identity and `coverCanonicalPivotal` returns a scalar with no `PivotalStructure`.

5.2 **The torsor.** Given one pivotal structure, the set of pivotal structures is a torsor under $\mathrm{Hom}(U_{\mathcal{C}}, \Bbbk^\times)$, where $U_{\mathcal{C}}$ is the universal grading group. This is a statement about *differences* between existing structures.

5.3 **What is withdrawn.** The draft claimed the criterion

$$\mathrm{Pivotal}(\mathcal{C}) \neq \varnothing \iff [g] \text{ is a square in } \mathrm{Hom}(U_{\mathcal{C}}, \Bbbk^\times),$$

and the corollary that $\mathcal{C}$ is pivotal when $U_{\mathcal{C}}$ has no 2-torsion. Neither is available. A map $g\colon \mathrm{Id} \to (-)^{****}$ is not an automorphism of Id, so its class is not a character class without an identification the project does not have; and the criterion fails even for monoidal g: $\mathrm{Vec}_{\mathbb{Z}/2}$ with g the non-trivial character is pivotal, while that character has no square root in $\mathrm{Hom}(\mathbb{Z}/2, \mathbb{C})$. A torsor does not establish existence. Any replacement needs a precise source and a separate proof, and is not promised.

Deliverables. D7. **Success criterion.** The cover is a `FusionSystem` with a `PivotalStructure` and is described as an equivariantization; the torsor statement is proved for differences; no existence criterion is stated.

WP6 — The categorical bridge and upstream (months 12–24)

Objective. Mathlib's categorical definitions become the ones the library uses, and the vendored placeholders disappear.

Tasks.

6.1 **The pivotal equivalence.** On the category of WP2, prove PivotalStructure $S \simeq$ PivotalCategory (CategoryOf S): Mathlib's monoidal natural isomorphism $\mathrm{Id} \to (-)^{**}$ restricts to the scalars p_i and `p\_mon` is its coherence.

6.2 **Delete the placeholders.** When #42145/#42150/#42191 merge, replace `PivotalCategory.lean` by an import. Until then the file declares nothing used elsewhere, and the blueprint says so.

6.3 **Contribute back.** The Perron–Frobenius material and the spherical layer are the two pieces of independent upstream value.

Deliverables. D8. **Success criterion.** `MathlibPatches/` contains no mathematics, only imports.

WP7 — Instances, witnesses, and the regression suite (continuous)

Objective. Every theorem in the library has an inhabitant, and every defect in the table of §4.2 has a regression test.

Tasks.

7.1 Extend the instance set, over $\mathbb{C}$ as well as $\mathbb{R}$: $\mathrm{Vec}^{\omega}_{\mathbb{Z}/n}$ for $n = 2, \ldots, 6$ with genuine cocycles; Tambara–Yamagami for $\mathbb{Z}/2$; the rank-three ring $K(1, 0, 1)$ already present; H_3 (non-integer FPdim) as a test that nothing assumes integrality.

7.2 Formalise the two script-level witnesses of §4.5: $\mathrm{Vec}^{\omega}_{(\mathbb{Z}/2)^2}$ with $J \equiv 1$ (the failure of (M)) and $\mathrm{Vec}^{\omega}_{(\mathbb{Z}/3)^2}$ (the failure of $J^2 = 1$).

7.3 A *negative* multiplicity test. The based ring $X^2 = \mathbf{1} + 2X$ is legitimate, but it has no fusion-category categorification over $\mathbb{C}$ ([Ost02], Theorem 2.1). It is used to test the indexing of a matrix-valued associator and as a negative example against any claim that a based ring categorifies; it cannot supply a coherent rigid witness, and no such witness for $N = 2$ is promised by this programme.

7.4 A census harness with genuine cocycles, failing the build on disagreement with the reference computations.

Deliverables. D9. **Success criterion.** Sixteen data sets, at most two of them script-only, at least four with genuinely non-trivial cocycles, and the census harness green.

5.4 Milestones

Milestone	Month	Content	Deliverable
M0	2	First delegation batch passed review: reproduced clean build; enforcing audit gate green with its self-test; census repaired; signature and exact target settled (statements and choices recorded; proofs and re-verification due at M2/D3)	D2 (part)
M1	4	`FPData` proved or labelled; clean-tree CI with the enforcing audit gate; blueprint with the statement-to-source table	D1, D2
M2	8	Signature v2 (the $\mathbb{C}$ scalar contract, triangle derived, qualified coherence, gauge fix); *exact* target; six instances re-verified	D3
M3	10	Reconstruction and trace correspondence; `derivedJ` defined as the tensorator morphism	D4
M4	14	J derived; four laws and qualified coherence proved; census repaired and kernel-checked	D5, D11
M5	18	a constructed; T-matrix domination; bound $\lvert d_i\rvert \le f_i$; sum identity and equality case	D6 (part), D9
M6	20	σ (tensor) and τ (natural) normalisations; `ENO823Exact` proved (over $\mathbb{C}$, derived twist / `J_match` in the antecedent)	D6
M7	22	Cover as equivariantization; torsor for differences; withdrawn criterion recorded	D7
M8	24	Bridge to `PivotalCategory`; placeholders deleted; release v1.0; papers submitted	D8, D10–D12

6 Techniques

6.1 Finite data and exact arithmetic

Everything categorical is reduced to finite algebra over a field, and everything numeric is done exactly. Two devices carry the instances.

Integer pairs in $\mathbb{Z}[\sqrt{d}]$. Real numbers never appear in a proof obligation. Each entry of F, ev, coev, J in the Fibonacci, Ising and Yang–Lee examples is an integer pair (p, q) denoting $(p + q\sqrt{d})/2$ in $\mathbb{Q}(\sqrt{d})$; the field operations are defined on pairs, and `decide` checks the pentagon, the unit laws, the inverse laws and the snakes on all label tuples. For Ising, $d = 2$ and the scalar $1/\sqrt{2}$ enters as a normalisation, so the identities are checked twice — once for the rational part and once for the $\sqrt{2}$ part. This is why the examples are trustworthy despite being numerical, and it is why they compile in 100 seconds rather than hours.

Field checks are hand-written. Closing real-number arithmetic is not automatic: `one_lt_pow` and its primed variant were unusable in the Yang–Lee file, and the φ^4 contradiction had to be assembled by hand from `nlinarith` with an explicit $\sqrt{5} \cdot \sqrt{5} = 5$. A mis-stated closed form for $((1 + \sqrt{5})/2)^2$ cost a build cycle. The practical rule adopted is: get the closed form right on paper, then prove it once, then reuse.

6.2 Inhabitant-first statement design

The four defects of §4 share one cause: a statement was written to look like its source and then trusted. The remedy, now institutionalised in this project, is that *no statement is committed until*

an inhabitant exists, and no axiom is believed until an example satisfies it and a non-example does not. Concretely:

(i) every structure gets a witness in the same commit as its definition (`VecZ2.system` with the definition of `FusionSystem`);

(ii) every axiom gets a regression test that fails if the axiom is strengthened (`swapped_Finv_comp_ne_one` for the snake; `VecZ2Twisted` for the twist);

(iii) every theorem whose statement mentions a hypothesis gets a witness that the hypothesis is needed (Yang–Lee for positivity; $\mathrm{Vec}^{\omega}_{(\mathbb{Z}/2)^2}$ for multiplicativity).

The cost is real — the witness files are about half the library — and it is the single highest-value investment in the programme, because the defects it catches are exactly the ones that survive refereeing.

6.3 Falsification before proof

For each proposed statement S, the programme first looks for a counterexample, in this order: (i) on paper, on the smallest rank; (ii) in Python, on the full data set, with the associators composed as matrices rather than manipulated symbolically; (iii) in Lean, by attempting the statement on the counterexample candidate before attempting the theorem.

This order is not a heuristic. Two of the project's four defects were found this way, and a third — the '$J \equiv 1$' claim for pointed categories — was refuted by an exhaustive computation in `FusionCategory/docs/pointed_twist_check.py` after it had been stated in the plan. The Python step matters because matrix composition over channel bases is where index conventions break, and because a script that recomputes every scalar from the snake axioms is an independent implementation, not a restatement.

6.4 Multiplicities: why this is a separate development

With $N_{ij}^k > 1$ the Hom space $\mathrm{Hom}(X_k, X_i \otimes X_j)$ has dimension N_{ij}^k and the associator at fixed (a, b, c, d) is a matrix whose entries are themselves matrices; the pentagon becomes a matrix identity with channel bases on both sides, and "the" F-symbol is a choice of bases. Three consequences:

(i) the scalar signature cannot be extended by adding a parameter — the type of `F` changes, and every instance proof that ends in `decide` over `Fin n` must be rewritten;

(ii) gauge freedom is larger (a GL_N-valued change of channel bases); the twist J becomes an operator between multiplicity spaces, and the condition it satisfies — multiplicativity, or the per-channel identity — is an *operator equality on each multiplicity space*, not a scalar equation. Taking a determinant or a trace of that equality destroys the monoidality information and does not give the scalar law: the general statement has no scalar reformulation of the kind the multiplicity-free case has;

(iii) the correct target in the general case is the *categorical* statement (WP6), with the skeletal statement as a corollary in the multiplicity-free case.

The proposal therefore treats the multiplicity-free theorem as the deliverable and the general case as a stretch goal, pursued through the categorical bridge rather than through a bigger scalar signature. This is a scope decision, stated here so that it can be argued with.

6.5 Diagram evaluation and independent reference computations

The one place where the project has repeatedly gone wrong is the evaluation of a diagram with more than two F-moves. The technique that works is to evaluate the same diagram twice by different routes:

- an *evaluation-side* assembly (compose evaluations, transport coevaluations);

- a *coevaluation-side* assembly;
- the Ω-coefficient form of WP2 Task 2.2;
- the direct φ-evaluation from its defining property $\mathrm{ev} \circ (\varphi \otimes \mathrm{id}) = \mathrm{ev}$, with full associator bookkeeping (`FusionCategory/docs/honest_twist_check.py`).

Agreement of four independent assemblies on sixteen data sets is the current standard of evidence for a formula, and it is what caught the defective transcription of Ω recorded in the plan. It is *not* a proof, and the proposal does not treat it as one: the kernel proof of WP2 Task 2.3 is what settles the formula, and the scripts are demoted to regression tests.

A second, independent technique is the Temperley–Lieb check: in $\mathrm{TL}(\varphi)$ one can compute the loop scalar of Fibonacci with no F-data at all, and it gives $\kappa_\tau = \varphi$ against the naive φ^2. It certifies the snake normalisation and nothing else, which is exactly the scope claimed for it.

6.6 Parallel adversarial exploration

The project uses a workbench in which several agents attack one problem in independent directions, register their direction on a shared board, and submit reports that a coordinating agent accepts only after re-deriving or kernel-checking the claim. Two things are worth recording, because both bear on the proposed programme.

What it bought. The pointed-case value of the twist, $J = \delta(\kappa) = \kappa_a\kappa_b/\kappa_c$, was settled in this mode by three independent assemblies; the identification of the trace identity with (M) emerged from it; and four independent evaluations of the J-diagram agreed, which is what isolated the defective Ω transcription.

What it got wrong. The same mode produced claims that did not survive checking: that $X^{**} \cong X$ canonically in any rigid monoidal category (false — this is precisely pivotality, and adopting it would have made the signature vacuous, as it had already done once); a scalar formula for ψ that returns 1 on channels where the honest twist is -1; and the assertion $J \equiv 1$ for pointed categories, which is false and was replaced by $\delta(\kappa)$. Each was caught by the acceptance rule, not by the generating agents.

Consequence for the programme. The technique is used for exploration and for independent assembly, never as evidence. Every claim in §4 marked "kernel" has been re-proved or re-checked by the coordinating agent; claims marked "script" are labelled as such; and no report is merged on the strength of its own text. The proposal's WP3 and WP4 explicitly budget for this: parallel exploration of the positivity step and of the cover, with kernel acceptance.

6.7 Statement fidelity as a method

The four defects are instances of four general failure modes, and naming them is part of the proposed contribution.

F1. Vacuity. The hypothesis is inconsistent, so the theorem is true and empty. Detected by inhabitation, not by proof.

F2. Drift. The formal statement is a different, usually stronger, statement (a one-line axiom where the source has a coherence condition). Detected by proving the source's examples fail, or by checking that the statement is not provable without its hypothesis.

F3. Unrepresentability. The signature cannot express the general case (real scalars where the source is complex; no multiplicity). Detected by trying to state the general theorem, not by proving the special one. Note that the label question is *not* an instance: double duality always fixes isomorphism classes (ENO Proposition 2.1), and an earlier draft of this proposal wrongly listed $i^{**} \neq i$ here.

F4. Normalisation. The statement is right but the gauge is wrong: a reciprocal where a matrix inverse is required, a sign convention that excludes the examples. Detected by the examples, and only by the examples.

The audit protocol that follows is mechanical and is written up in Appendix B. It has three gates: *inhabitation* (does the structure have a witness?), *sharpness* (is the hypothesis used, i.e. does dropping it admit a counterexample?), and *correspondence* (is the statement the source's, with its numbering and its quantifiers?). The third gate is the one a proof assistant cannot automate, and the one that caught D3.

6.8 Engineering discipline

- **Pinned toolchain.** Lean `v4.33.1`, Mathlib `0df444a3`, in both packages; no bump without updating both lockfiles.
- **No `sorry`, no `axiom`**, axiom budget `[propext, Classical.choice, Quot.sound]` enforced by the audit gate of Gate 4 (Appendix B) — a script that fails the build on Lean errors and on out-of-budget dependencies, self-tested against a deliberate `sorry`. The printed `AxiomAudit.lean` report alone enforces nothing.
- **Per-file builds** with generous timeouts (an `import Mathlib` costs 20–60 seconds; the example files cost minutes), `weakLeanArgs = ["-j2","-s8192"]` to stay inside CI memory limits.
- **No module-private declarations**: the project avoids the bare `module` header, which makes declarations private and breaks downstream imports in a way that is hard to diagnose.
- **Commit messages as a proof log**, including retractions. The repository's history carries three commits whose subject is a correction of the previous commit's mathematics; that is a feature, and the plan keeps the record rather than rewriting it.

7 Risks and mitigations

The risk of this programme is not concentrated in the analysis. It is concentrated in *statement correctness* and *reconstruction*: the preliminary work got four statements wrong before getting them right, and the analytic step cannot even be stated until the skeletal datum is a category. The table makes the dependencies explicit, in the order in which they bite.

	Risk	Mitigation / fallback	Level
R1	**Statement correctness.** A work package targets a statement that is not the source's, or a normalisation that is wrong. Four instances already occurred in the preliminary work, and an external reading of the first draft of this proposal found six more.	Gate 3 of Appendix B: every theorem carries its source, edition, numbering and quantifiers; the exact target is reproduced verbatim; two independent readings of each new statement before it is scheduled.	high
R2	**Reconstruction.** The skeletal datum may not reconstruct as a fusion category from the present axioms, and the reconstruction is real category theory inside Mathlib.	WP2 does the multiplicity-free case first and is scheduled early; fallback: every theorem is labelled "skeletal" rather than "categorical" and the categorical claim is withdrawn from the abstract.	med/high

	Risk	Mitigation / fallback	Level
R3	**The twist derivation.** No correct closed form is known; the Ω formula in `PLAN.md` is retracted, and the census that appeared to support it is inconclusive.	Derive through the comparison ψ, kernel-check, and cross-check against Bartlett's Theorem 4.17 and Corollary 3.26; fallback: keep J a hypothesis and state the theorem conditionally on it.	med/high
R4	**The quadruple dual.** ENO Theorem 2.6 needs weak Hopf algebras, out of scope; Bartlett's graphical proof is the alternative and is a large formalisation.	Budgeted as a separate task with its own deliverable; fallback is the conditional statement, labelled.	med
R5	**Complex scalars.** The library is over $\mathbb{R}$; $\mathbb{C}$ breaks the `decide`-based instance proofs, which use integer pairs in $\mathbb{Z}[\sqrt{d}]$.	Model $\mathbb{Q}(\zeta)$ by integer polynomials mod Φ_n and keep `decide` exact; keep the $\mathbb{R}$ instances and add $\mathbb{C}$ ones rather than migrating.	med
R6	**The normalisation step.** ENO's σ/τ argument needs the operator-level multiplicativity that WP2 supplies; if the reconstruction slips, this is unavailable.	The conditional theorem `eno823_of_pseudoUnitary` already isolates the gap; publish the conditional result if necessary.	med
R7	Mathlib churn: #42150/#42191 merge with different names, or stall.	Vendor, keep byte-compatible, switch by replacing one import; contribute the spherical layer if it stalls.	low
R8	Elaboration cost: `decide` over 3^n label tuples grows fast; example files already cost 100 s.	Exact arithmetic in $\mathbb{Z}[\sqrt{d}]$; per-file builds; abstraction barriers so the core does not import the examples; `native_decide` forbidden.	low
R9	**Scope.** A single investigator over 24 months, on a theorem whose unconditional form exceeds the plan.	The scope statement of §2.6 is part of every deliverable; D1–D6 are the core and D7–D12 are additive; WP5 is off the critical path.	med

The stop conditions. Two, and they are decided at fixed months rather than at the end. If by month 12 the reconstruction of WP2 has not produced a category, the programme drops the categorical claim and publishes the skeletal theorem as such. If by month 16 the twist derivation of WP3 has not succeeded, the theorem is stated with J as an explicit hypothesis, and the deliverable is the conditional result together with the Yang–Lee sharpness theorem. Neither outcome is a failure; both must be decided on time.

8 Timeline

Twenty-four months, eight quarters. Bars mark active work. The order is forced by the dependencies: signature and exact target (WP1), then the reconstruction that makes the words mean something (WP2), then the twist (WP3), then ENO's normalisation (WP4). The cover and grading work (WP5) is deliberately late and off the critical path, because its statements are the ones most in need of justification.

Work package	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8
WP0 Foundations	■	■						
WP1 Signature & exact target	■	■	■					
WP2 Reconstruction & trace	■	■	■	■				
WP3 Derived twist		■	■	■	■			
WP4 ENO normalisation & theorem			■	■	■	■	■	
WP5 Cover & grading					■	■	■	■
WP6 Bridge & upstream					■	■	■	■
WP7 Instances & tests	■	■	■	■	■	■	■	■
Milestones	M0	M1	M2	M3	M4	M5	M6	M7, M8

9 Expected deliverables

Twelve deliverables, each with a month and an acceptance criterion. D1–D6 are the core; D7–D12 are additive and independently valuable. The scope labels (S1)–(S4) of §2.6 apply to every deliverable below, and the conditional and unconditional forms of the theorem are listed separately.

	Deliverable	Acceptance criterion	Month
D1	**Core library v1.0**: fusion rings; Perron–Frobenius dimension with `FPData` proved or explicitly labelled; skeletal systems; pivotal and spherical layers; the twist reduction; ENO 8.21.	Reproducible clean build; no `sorry`; every audited declaration at `[propext, Classical.choice, Quot.sound]`.	4–20
D2	**Reproducibility kit**: pinned toolchain, CI gates, clean-tree script, per-file harness, archived artifact with a DOI.	A fresh clone builds and audits without manual steps; the difference from the working-tree build is documented.	4
D3	**Corrected signature and exact target**: the $\mathbb{C}$ scalar contract; triangle law derived; coherence with the nonzero-entry qualification (explicit nonzero side condition or retained F-factors); gauge-invariant duality normalisation; the target matching ENO 8.23 / EGNO 9.5.1 field by field.	All six existing instances re-verified; the new lemmas proved, not assumed; the target's existential conjunction reproduced verbatim, stated over $\mathbb{C}$ with FP dimensions embedded from $\mathbb{R}$.	8
D4	**Reconstruction and trace correspondence**: the multiplicity-free datum is a fusion category; every scalar in the target is the corresponding categorical trace; `derivedJ` defined as the tensorator morphism, independently of the carried field.	The reconstructed category satisfies the `IsFusionCategory` predicate of Task 2.1; the trace identities are theorems; the carried-versus-derived relation is `J_match` or a field replacement.	10
D5	**Derived twist**: the comparison ψ, the ratio formula, the four laws, the qualified coherence, a repaired census with genuine cocycles, and the Bartlett comparison.	No hypothesis on `derivedJ` beyond the four laws — a carried J, where one remains, linked by `J_match` proved per instance; at least four genuinely non-trivial cocycles, kernel-checked; the retracted Ω either corrected or replaced in `PLAN.md`.	14

	Deliverable	Acceptance criterion	Month
D6	**ENO 8.23**: a constructed, T-matrix domination, the bound $\lvert d_i\rvert \le f_i$, the sum identity, the σ/τ normalisation, uniqueness.	`ENO823Exact` proved for multiplicity-free data over $\mathbb{C}$ under the scalar contract of WP1 Task 1.1, with the derived twist (or `J_match`) in the antecedent per Task 1.5; the hypothesis on g either discharged (Bartlett) or explicitly labelled.	20
D7	**Cover and torsor**: the cover as an equivariantization, with doubled dimension; the torsor for differences of pivotal structures; the withdrawn existence criterion recorded.	The cover is a `FusionSystem` with a `PivotalStructure`; no existence criterion is claimed.	22
D8	**Mathlib contributions**: the pivotal/spherical bridge; Perron–Frobenius dimension; the vendored placeholders deleted.	`MathlibPatches/` contains only imports.	24
D9	**Instance and regression suite**: sixteen data sets over $\mathbb{R}$ and $\mathbb{C}$, at least four with genuinely non-trivial cocycles; the two script-level witnesses formalised; the rank-two negative test.	Census harness green; one regression test per defect D1–D4.	18
D10	**Papers**: (a) the formalisation of ENO 8.23, with its scope in the title; (b) a negative-results note ("ENO 8.23 without pseudo-unitarity is false", with the kernel-checked Yang–Lee witness); (c) a methodology note on statement fidelity, including the corrections this revision records.	Submitted: (b) by month 12, (a) by month 21, (c) by month 24.	12–24
D11	**Blueprint and documentation**: statement-to-source table, module dependency graph, gauge conventions, scope labels, a worked tutorial on adding an instance.	A reader can add a new instance without reading the core, and can tell which theorems are conditional.	12
D12	**Open-problem dossier**: formal statements of ENO Conjecture 2.8 and its variants, with the known partial results and the constraints any counterexample must satisfy.	Reviewed against the literature; every claim labelled proved/checked/conjectured/resolved.	24

What is not a deliverable. A proof of ENO Conjecture 2.8; the weak Hopf algebra input (ENO §4–5, Theorem 2.10) unless the Bartlett route succeeds; the general multiplicity case; any analytic statement over a general field — the absolute values and positivity of the theorem are $\mathbb{C}$-specific (S2 of §2.6); and the classification of rank-three pivotal categories beyond the inequality used as a test. These are out of scope and are stated as such in the blueprint, so that a reader cannot mistake the boundary of the result.

10 Significance

For the classification programme. Proposition 8.23 is used as a black box in a large and growing literature; a kernel-checked proof would let downstream work cite a machine-checked theorem instead of a two-page argument whose normalisations are easy to get wrong. The present project's own history is the evidence: four defects, two of them in axioms, one of them turning the target into a tautology. A reader who trusts the literature's normalisations is trusting something that a careful formaliser, working from the source, got wrong four times.

For the mathematics. The formalisation has already produced statements the literature leaves implicit: that (K), the per-channel identity, is *not* invariant under the vertex gauge and survives only the duality normalisation $\lambda_{i^*} = \lambda_i$; that (C) and (M) are necessary but not sufficient in the signature, and that both are satisfied by data on which the target fails; that the trace identity of ENO's proof is exactly (M); and that the coherence law needs the nonzero-entry qualification of Bartlett's Corollary 3.26. None of these is a new theorem of fusion categories in the sense of a new mathematical result, but each is a precision about hypotheses that the informal literature elides, and precision of this kind is what makes a black box safe to use.

For Mathlib and the formalisation community. The categorical layer is arriving now (#42145, #42150, #42191, #42192). What is missing after it lands is exactly what this proposal builds: the finite-data layer, the examples, the counterexamples, and one substantial theorem that exercises the layer. The Perron–Frobenius material and the spherical layer are independent upstream contributions; the sixteen verified data sets are the regression suite that the categorical definitions will need.

For the author's programme. The sibling Haagerup–Izumi development formalises the analytic side of a classification theorem (coefficient identities, reduction to finitely many algebraic equations). This proposal formalises the algebraic side of the same story (dimensions, pivotal structures, sphericity). Together they cover the two halves of a verified pipeline from fusion rules to a categorical structure, and the Haagerup–Izumi categories are the natural large-scale instance for WP6.

Broader impact. Verified anyon data — F and R symbols with a checked pentagon and checked pivotal structure — is the input to proposals for topological quantum computation, and the failure mode there is precisely an unchecked normalisation. The library does not build a quantum computer, but it does make the statement "these symbols satisfy the axioms" a machine-checked one, and that is the part which is currently done by hand.

Limitations, stated plainly. The theorem is formalised for multiplicity-free data in a skeleton; the general case is a stretch goal through the categorical bridge. The analytic input of ENO is formalised, but its weak Hopf algebra prerequisite is taken as a hypothesis. And the work is a formalisation: its novelty is in precision, verification, and counterexample, not in a new route through the mathematics.

Acknowledgements

The preliminary results reported here were developed in the `fusion-category-lean` repository, whose `PLAN.md` records both the results and the retractions; the retractions are part of the evidence and are kept. The Haagerup–Izumi development in the sibling package is the work on which the feasibility argument rests.

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A Module inventory and dependency graph

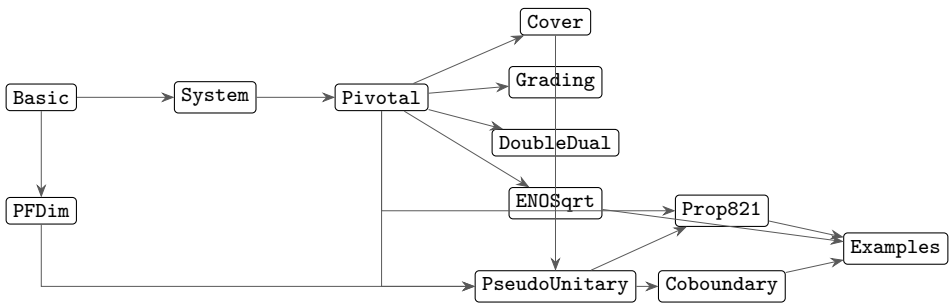


Figure 2: The import spine of the library (transitive imports suppressed; `Sanity`, `Step4Checks` and the individual `Examples/` files are leaves off this spine). `Examples/` is the umbrella over Fibonacci, Ising, Ostrik, $\mathrm{Vec}^{\omega}_{\mathbb{Z}/2}$, Fibonacci–Ising and the step-4 checks; Yang–Lee and the pseudo-unitary instances are imported separately by the audit.

Module	Lines	Decls†	Content and status
`Basic`	229	26	fusion ring, fusion/FP matrices, `IsPFEigenvector`, `FPData`, global dimension. Complete; `FPData` is a hypothesis.
`System`	468	19	`FusionSystem`: F, `Finv`, snakes, J (retained-F coherence), `ev_dual`; $\mathrm{Vec}_{\mathbb{Z}/2}$ inhabitant with the 512-tuple pentagon. Complete.
`Pivotal`	191	21	`PivotalStructure`, `dim`, `IsSpherical`, `isSpherical_iff`, `pVal_dual`. Complete; the nontrivial-J witness is owed (batch item B3, Gate 1).
`PFDim`	49	5	`PFUnique`, φ and its uniqueness. Complete (conditional on `FPData`).
`Cover`	128	8	`S4Data`, `CoverObj`, forgetful functor, `coverGlobalDim_double`. Scaffolding: no `PivotalStructure` on the cover yet.
`Grading`	153	8	universal grading, `pVal_dual`, two `def`s recording open targets. Formerly three axioms; now axiom-free.
`DoubleDual`	155	9	the older derived twist in the strict self-dual normalisation. Lags the ψ-derivation; to be rewritten against `Finv`.
`Coboundary`	334	15	`twistNorm`, `dim_eq_twistNorm`, `eno823_of_twistNorm` and its converse; the pointed `KappaMultiplicative` layer retained for reference. Complete.
`Prop821`	474	21	cyclic symmetry, `fpdim_char`, the weighted-norm bound, the equality case, `tMatrix`, `mult_iff_twist`. Complete modulo `DimEigen`.
`PseudoUnitary`	459	37	`ENO823`, `ENO823Exact`, `IsPseudoUnitary`, `IsPseudoUnitaryPivotal`, `eno823_of_pseudoUnitary`, `pivotal_ext`. Complete as a conditional reduction.
`ENOSqrt`	232	13	`SqrtData`, b, `Tmat`, d, `b_sq`, `trace_iff`, `toPivotal`. Complete; existence of a is assumed.
`Sanity`, `Step4Checks`	136, 147	8	numerical and consistency checks. Complete; imported by the umbrella.
`Examples/FibonacciIsing`	953	115	Fibonacci and Ising over $\mathbb{R}$: every field proved, spherical, dim = FPdim. Complete; $\sim$100 s.
`Examples/YangLee`	420	39	the refutation of the un-hypothesised target. Complete.
`Examples/VecZ2Twisted`	89	12	two pivotal structures, $\dim g = \pm 1$; pseudo-unitary (the corrected doc). Complete.
`Examples/Ostrik`	154	21	the rank-three ring $K(1,0,1)$ and its `FPData`. Complete.
`Examples/`$\cdots$	438	38	Fibonacci, Ising, pseudo-unitary instances, $Z/3$ note.
`Examples/Z2SqFalseJ`	684	76	the $(\mathbb{Z}/2)^2$ false-J witness: `not_eno823`, `not_eno823Exact` under satisfied pseudo-unitarity, (M)/(K) failures. Complete.
`Examples/Z2SqHonest`	1273	135	the honest-twist witness: $J = \delta(\kappa)$, the nonconstant pivotal $p = 1/\kappa$ (Gate 1's witness), spherical with dim = FPdim; the positive instance of the corrected target; the forcing theorems (dim $\equiv 1$ forces $J = \delta(\kappa)$, so `J_match` is a consequence on this datum). Complete.
`Witnesses`	486	90	the $(\mathbb{Z}/3)^2$ coboundary certificate (`om_is_coboundary`), the genuinely twisted family (`no_trivialising_cochain`, J^2 a primitive cube root), and the D_8 row. Complete.
`Triangle`	89	3	the triangle law from the pentagon and the 1×1 block + `Finv_comp`. Complete.
`Coherence`	54	6	the retained-F and qualified coherence forms, both directions. Complete.
`Statements`	287	21	the frozen $\mathbb{C}$ statements, the four decisions, the two collapse guards. Complete.
`MathlibPatches`	55	0	stubs for #42150 and the PF chain. Stub: WP0/WP5.
Total	**8,135**	**745**	31 content files (plus 3 umbrellas and the generated audit); clean build reproduced (B1); audit 1,058 declarations (715 theorem-kind), mechanically enumerated.

†The Decls column is a keyword grep (the honest source-level count; the conformance audit found the v1.2 figure of 414 was this grep with one hand correction). The *audited* declaration count — every exported constant, including compiler-generated lemmas and structure projections, enumerated mechanically from the elaborated environment — is 1,058, and that is the number the gate covers.

B The audit protocol

The protocol is a sequence of gates applied to every declaration before it is merged. Its purpose is to catch the failure modes F1–F4 of §6.7, none of which is caught by a successful build.

Gate 1 — Inhabitation. Every structure introduced gets a witness in the same commit, and the witness is a case from the literature or a standard construction, not a degenerate one. For `FusionSystem` the witness is $\mathrm{Vec}_{\mathbb{Z}/2}$; for `PivotalStructure` the witnesses are $p \equiv 1$ on a datum with $J \equiv 1$ (Fibonacci, Ising, $\mathrm{Vec}_{\mathbb{Z}/2}$) and a *nonconstant* p on a datum with nontrivial J: the pointed $\mathrm{Vec}^{\omega}_{(\mathbb{Z}/2)^2}$ datum with the honest twist $J = \delta(\kappa)$ and $p_i = 1/\kappa_i$, spherical with $\dim \equiv 1 = \mathrm{FPdim}$. A constant p cannot witness a datum whose J is nontrivial *on an active channel* — through `p_mon`, $p \equiv 1$ forces $J = 1$ on every channel with $N_{ij}^k > 0$, while off the active channels J is unconstrained by every axiom, so a constant p can accompany an off-channel-perturbed J and the witness must be non-degenerate — so the pairing an earlier draft of this gate requested is contradictory on the active channels. Until the $(\mathbb{Z}/2)^2$ witness is kernel-checked (WP3 Task 3.4) the nontrivial-J slot of this gate is unfilled: the pivotal layer is merged with a debt, its nontrivial-J witness being due at B3/M0. A structure with no witness is not merged.

Gate 2 — Sharpness. Every hypothesis is either used essentially or is recorded as unused. A hypothesis H is *used essentially* if dropping it makes the statement false on a witness. The two witnesses carried for this purpose are Yang–Lee (dropping pseudo-unitarity) and $\mathrm{Vec}^{\omega}_{(\mathbb{Z}/2)^2}$ (dropping multiplicativity). Where no counterexample is known, the declaration says so.

Gate 3 — Correspondence. Each theorem is entered in the statement-to-source table with (i) the source, (ii) the source's numbering in the edition used, (iii) the quantifiers and the carrier, (iv) any hypothesis of the source that is absent and why. The ENO numbering trap (8.23 in the Annals, 8.22 on the arXiv) is recorded there. This is the gate a proof assistant cannot automate, and the only one that catches drift.

Gate 4 — Axiom audit, enforcing. `AxiomAudit.lean` prints the axioms of the declarations it lists, but printing is a *report*, not a gate: the earlier instruction — run the print and `grep` its output — could not fail on a Lean error or on an out-of-budget dependency, and a non-zero exit from `grep -v` usually meant *success*. The gate is the checked-in script `FusionCategory/tools/audit-gate.sh`, with the checked-in manifest of audited names `tools/audit-manifest.txt` and a self-test under `tools/selftest/`. Its contract: exit non-zero if (i) `lake env lean AxiomAudit.lean` exits non-zero, or its output contains a Lean error marker (`error:` / `error(`; a declaration *name* containing "error" does not match the marker); (ii) any audited declaration depends on an axiom outside `[propext, Classical.choice, Quot.sound]` — strict subsets are within budget; a `sorry` appears as `sorryAx`, and an `axiom` declaration by listing its own name, which this branch rejects (`#print axioms` applied to an `axiom` does *not* error on the pinned toolchain, it prints the axiom's own name; a name whose module is not imported produces an `Unknown constant` error line, caught by (i); both were mistaken for real findings in this project); (iii) any output line is not a recognised audit line — recognised are the two `#print axioms` forms with wrapped lines rejoined before matching, blank lines, and `warning:` lines (a `sorry` is caught by (ii) via `sorryAx`, not by its warning line; trailing text after the bracket list is rejected); or (iv) the output

is empty, or its set of declaration names differs from the manifest — so deleting or editing a `\#print axioms` line fails the gate. Generating the audit list mechanically (every declaration, not a hand-maintained list) is batch item B2, and until then coverage is exactly the manifest. The gate reports every violation before failing.

The self-test is checked in: one fixture per failure mode under `tools/selftest/bad/` (a `sorryAx` dependency, a declared `axiom`, a Lean error line, junk, trailing text after the bracket, a name containing "error", a wrapped out-of-budget line, and an audit with a line missing against the manifest), plus the passing cases under `tools/selftest/good/` (in-budget, strict subsets, wrapped compliant lines, and a frozen snapshot of the real audit output checked against the real manifest); `tools/selftest/run.sh` asserts the gate fails on every bad fixture and passes on every good one, and is a CI gate itself. A gate that has never been observed to fail is a report.

Gate 5 — Build hygiene. No `sorry`, no `axiom`, no bare `module` header, warnings permitted. The full build must pass from a clean tree (WP0 Task 0.2), not merely incrementally.

Gate 6 — Regression. Each defect of Table 1 has a named test that fails if the defect returns: `VecZ2.system` (D1), the 512-tuple pentagon (D2), `VecZ2Twisted.system` (D3), `swapped_Finv_comp_ne_one` (D4).

C The main statements

For reference, the target statements in mathematical form, with their Lean names.

Pseudo-unitarity. `IsPseudoUnitary S h` is the scalar equation

$$\sum_i \kappa_i \, \kappa_{i^*} \;=\; \sum_i f_i^2,$$

with no pivotal structure in it.

The bare target. `ENO823 S h` is

$$\exists\, P : \text{PivotalStructure } S, \quad P.\text{IsSpherical} \;\wedge\; \forall i,\ \dim_P(i) = f_i.$$

The exact target. `ENO823Exact S h` states: for every pseudo-unitary S there is a unique pivotal structure P with

$$P.\text{IsSpherical}, \qquad \forall i,\ \dim_P i > 0, \qquad \forall i,\ \dim_P i = f_i.$$

The existence is inside the unique existential; a rendering that lists a spherical structure and then asserts positivity and uniqueness *conditionally* on $\dim_P = f$ is strictly weaker, and the earlier draft of this appendix did exactly that. It is withdrawn. In the $\mathbb{C}$ statement (WP1 Task 1.1), the conjunct $\dim_P i > 0$ is read through the positivity predicate of S2 — z real with $z > 0$ — and in both the $\mathbb{R}$ and $\mathbb{C}$ statements it follows from $\dim_P i = f_i$ with $f_i > 0$; it is kept because the source states it (Gate 3), not because it is logically independent. The library's current `ENO823Exact` is stated over $\mathbb{R}$; the $\mathbb{C}$ restatement is the same rendering under the scalar contract, with a real `FPData` embedded by $\mathbb{R} \to \mathbb{C}$. And the target as displayed quantifies over the *carried* J: the exact form takes the derived twist (or `J_match`) as a further antecedent, exactly as Task 1.5(b) records — without it, the statement is false on the pseudo-unitary $(\mathbb{Z}/2)^2$ datum with $J \equiv 1$.

The conditional form already proved. `IsPseudoUnitaryPivotal P` is the conjunction

$$(\forall i,\ \dim_P i > 0)\ \wedge\ \Big(\forall i,k,\ \sum_j N_{ij}^k \dim_P j = \dim_P i \cdot \dim_P k\Big),$$

and `eno823_of_pseudoUnitary` proves that these two conditions on one pivotal dimension, together with `PFUnique`, give `ENO823`.

ENO's input. `SqrtData` carries $a\colon \mathrm{Fin}\, n \to \Bbbk^\times$ with $a_i^2 = g_i$ and the monoidality of a^2 stated against J^2; then b, `Tmat` and d are

$$b_c^{ij} = J_c^{ij}\frac{a_i a_j}{a_c}, \qquad T_c^{ij} = N_c^{ij} b_c^{ij}, \qquad d_i = a_i \,\mathrm{coev}_i\, \mathrm{ev}_{i^*},$$

and `trace_iff` states that $d_i d_j = \sum_c T_c^{ij} d_c$ for all i,j is equivalent to (M) in the duality normalisation $\kappa_i' = \mathrm{coev}_i\, \mathrm{ev}_{i^*}$.

The derived twist. The object WP3 must construct is

$$J_{ab}^c \ = \ \frac{\psi_{a,b}|_c}{\psi_{b^*,a^*}|_{c^*}}, \qquad \psi_{a,b}\colon (a\otimes b)^* \to b^*\otimes a^*,$$

with ψ the unique evaluation-intertwining isomorphism. The closed form in terms of F and the restricted inverse block M_b^{-1} that `PLAN.md` displays is *retracted there* and is not quoted as a target; WP3 Task 3.2 derives the closed form and checks it against Bartlett's Theorem 4.17.

An ASCII rendering of the headline declaration. The Unicode binders are spelled out; this is a rendering for readability, not the source, and it reproduces the existential conjunction of `ENO823Exact` verbatim — the source's $\mathbb{R}$ is spelled R in this ASCII rendering, and the source's Fin n and Prop are unchanged.

```
-- the target, as the library actually defines it
def ENO823Exact (S : FusionSystem n R) (hFP : S.ring.FPData) : Prop :=
  IsPseudoUnitary S hFP ->
    exists! P : PivotalStructure S,
      P.IsSpherical /\
      (forall i, 0 < dim P i) /\
      (forall i, dim P i = hFP.fpdim i)
```

The conditional form `eno823_of_pseudoUnitary` replaces the antecedent by `IsPseudoUnitaryPivotal P` for a *given* P together with `PFUnique`; it is weaker than `ENO823Exact` precisely in that it assumes a pivotal structure rather than producing one.

D Changes in this revision

The first draft of this proposal was reviewed, and the review found eight substantive errors and a list of smaller ones. Each is recorded here with the change made, so that the revision can be checked against the review rather than against itself.

	Finding	**Change**
1	The nontrivial double-dual permutation is based on an error: every simple is isomorphic to its double dual (ENO Prop. 2.1); pivotality is about the monoidal structure.	The permutation is removed. Remark 2.6 now separates the objectwise statement from the monoidal one; WP1 records the label question as closed; failure mode F3 no longer cites it.
2	The promised unconditional theorem exceeds the stated scope (multiplicities, the quadruple-dual theorem, complex scalars).	§2.6 states (S1)–(S4); g is a hypothesis or Bartlett's Theorem 3.28; complex scalars are a WP1 task; the v1.1 abstract said "multiplicity-free, characteristic zero" (the v1.2 abstract states the $\mathbb{C}$ scalar contract instead).
3	The analytic package (old WP3, now WP4) misstates the phase correction: σ is a tensor automorphism of the identity, τ is an ordinary natural automorphism, and σ need not be sign-valued.	WP4 Task 4.5 rewritten with the two steps correctly typed, including the observation that a monoidal character leaves the defect b unchanged; §4 and §4.5 corrected.
4	The square-root criterion for pivotality is invalid, and the cover is an equivariantization.	WP5 Task 5.3 withdraws the criterion and the no-2-torsion corollary, with the $\mathrm{Vec}_{\mathbb{Z}/2}$ counterexample; the cover is restated as an equivariantization.
5	The displayed exact target omits existence.	Appendix C reproduces `ENO823Exact` verbatim, including the existential conjunction, and records the weaker rendering as withdrawn; WP1 Task 1.5 makes this a proof obligation.
6	$X^2 = \mathbf{1} + 2X$ has no categorification over $\mathbb{C}$; a determinant or trace does not make the general twist scalar.	WP7 Task 7.3 turns the ring into a negative test citing [Ost02]; §6.4 states that the condition is an operator equality on each multiplicity space, with no scalar reduction.
7	The twist package (old WP2, now WP3) reproduces a retracted Ω formula, and the cyclic census is vacuous.	WP3 opens with the retraction, takes the closed form as a task rather than an input, and requires four genuinely non-trivial cocycles, kernel-checked; the carry-cocycle bug is recorded.
8	The claim about the *number* of pivotal structures is overstated.	The abstract, Theorem 4.5 and its discussion now claim only that the *dimension function* is not determined by the ring, and note that the set of structures is a torsor once one exists.
a	The triangle axiom is derivable, not independent.	WP1 Task 1.2 proves $t = t^2 \Rightarrow t = 1$ from the pentagon and invertibility.
b	The twist coherence needs the nonzero-entry qualification.	WP1 Task 1.3 states it, citing Bartlett's Corollary 3.26, and derives the F-free form only where the cancelled entry is nonzero (v1.2: "only on the support" was the wrong notion — the support does not supply nonvanishing — and is corrected in the second-revision appendix, item 1).
c	(C)⇒(K) is not available in the signature.	Figure 1 redrawn: (C) and (M) are marked necessary and not sufficient, with the pointed false-J datum as the witness; the missing implication is attributed to the reconstruction and the twist.
d	$\mathrm{FPdim}(\mathcal{C}) = \sum_i f_i^2$, without the extra square.	Corrected in §2.2.
e	The pseudo-unitarity definition is circular.	Definition 2.7 now uses ENO's squared norms, available before pivotality, with the equivalence to (5) as a lemma.
f	The cyclic symmetry is missing a dual.	Corrected to $N_{ij}^k = N_{jk^*}^{i^*}$, the `cyclic1` statement.
g	EGNO attribute the theorem to Proposition 9.5.1.	Corrected in Theorem 2.8 and §3, with EGNO's own note identifying it with ENO 8.23.

h	Higher indicators are cyclotomic integers, not integral by sphericity.	Corrected in §3.

E Changes in the second revision (v1.2)

The revised proposal was reviewed a second time, and the review found five substantive defects and one structural instruction. Each is recorded here with the change made, so that this revision can be checked against the review rather than against itself.

	Finding	**Change**
1	WP1 mishandled zero associator entries: `F_support` and `J_ne_zero` were said to guarantee nonzero F on admissible channels. They do not — `F_support` forces zeros off the channels only, and `J_ne_zero` is a statement about J.	Task 1.3 rewritten: every cancellation carries an explicit nonzero side condition h_F, or the F-factors are retained in the law; no proof derives nonvanishing from the support axioms.
2	"A field of characteristic zero" cannot carry the analytic theorem: it supplies no absolute values, no square roots, no positivity; and a remaining reference derived positivity from a Hermitian structure.	The scalar contract (S2, Task 1.1): algebraic layer over a characteristic-zero field, analytic theorem over $\mathbb{C}$ only, with real `FPData` embedded, complex moduli, and positivity defined once; the Hermitian description removed from §1.5, `PLAN.md` and the Lean doc comments (`Coboundary.lean`, `PseudoUnitary.lean`).
3	WP2 Task 2.3 asked reconstruction to identify the carried field $S.J$ with the double-dual tensorator — an identification the false-J datum refutes.	Task 2.3 rewritten: reconstruction defines `derivedJ` independently; the relation to a carried J is the explicit hypothesis `J_match`, discharged per instance (Task 3.3), or the field is replaced by the derived definition.
4	Gate 1 requested $p \equiv 1$ on a nontrivial-J datum — contradictory, since `p_mon` then forces $J = 1$ on every active channel; and Gate 4's `\#print axioms` plus `grep` is a report that cannot fail on a Lean error or an out-of-budget dependency.	Gate 1 names a nonconstant witness ($p_i = 1/\kappa_i$ on the $(\mathbb{Z}/2)^2$ datum with $J = \delta(\kappa)$) and records the slot as unfilled until kernel-checked; Gate 4 is the enforcing script `tools/audit-gate.sh` with three failure conditions and a negative self-test.
5	Three contradictory instructions: `PLAN.md` still requested a nontrivial double-dual label permutation; Task 1.5 asked for an equivalence with a form it had just withdrawn as strictly weaker; Task 4.5 called the dimension ratios positive where only their squares are determined.	The request is withdrawn (ENO Prop. 2.1; `PLAN.md` now records the closure); Task 1.5 names its forms and both qualifications: exact ↔ bare *under* `IsPseudoUnitary` (bare → exact via `pivotal_ext` and FPdim positivity, exact → bare via `eno823_of_exact`), and the target itself stated for the derived twist or under `J_match`; Task 4.5 states the squared identity and defers the signs to the normalisation.
6	Structure: the review asked for a first delegation batch and an explicit source of truth before any unattended run.	§5.2: batch B1–B4 (reproduce the clean build, enforcing axiom audit, repair the cocycle census, settle the corrected signature and theorem statement), the source-of-truth paragraph superseding `PLAN.md`, and the holding rule that gates WP2–WP4 on the batch's review.

Post-audit corrections (v1.2.1). After v1.2 was committed, an eight-member audit — one direction per review point, plus cross-document consistency, Lean-source fidelity and an adversarial pass — found thirty-one further defects (consolidated; every one recorded with location, quote and minimal fix in the audit's reports), all fixed in this pass. The most substantial: the Hermitian-derived-positivity wording survived in `PLAN.md` and in two Lean doc comments (`Coboundary.lean`, `PseudoUnitary.lean`); `PLAN.md` asserted $\delta(\kappa) = 1$ on Fibonacci and Ising, which is false (φ^2 and 2 — not even legal twists there, since they violate `J_dual`) and was contradicted by a later bullet of the same file; the carried-versus-derived distinction did not reach the target statement itself, which is false on the pseudo-unitary $(\mathbb{Z}/2)^2$ datum with $J \equiv 1$ (Task 1.5(b)); the exact↔bare equivalence was stated unconditionally (Task 1.5(a)); the σ-step of Task 4.5 was not executable as written and is replaced by the one-rescaling normalisation; the enforcing audit gate passed on empty input, accepted trailing text after the bracket list, and rejected wrapped lines and names containing "error" — it is now checked in as `FusionCategory/tools/audit-gate.sh`, with a twelve-fixture self-test (every fixture verified to behave), a manifest of audited names, and a live run against the real audit output; and the module inventory, the timeline, and a set of cross-references were corrected. The gate's decision logic was validated independently by three executed harnesses during the audit.

The first delegation batch, landed (v1.2.2). The batch ran as an eight-member delegation (plus a three-member follow-up for the one open item), every report accepted only after the coordinator's own verification runs, and its artefacts are now in the tree: the false-J witness `Examples/Z2SqFalseJ.lean` (the target refuted under a satisfied pseudo-unitarity hypothesis — the kernel-checked justification for the `J_match` qualification); the census witnesses `Witnesses/` (the $(\mathbb{Z}/3)^2$ coboundary certificate and the genuinely twisted J^2 family); the frozen statements `Statements.lean` (with the two collapse guards: the `ENO823ExactC` family must be instantiated at Task 2.3's `derivedJ`, never universally quantified); the triangle law and the coherence equivalence (`Triangle.lean`, `Coherence.lean`); the corrected signature — `J_coherent` restated in the retained-F form and the `ev_dual` gauge normalisation added as a field, with all inhabitants unchanged (`rfl`); the mechanical audit (`tools/AuditList.lean` + `tools/gen-audit.sh`; 73 hand-picked names became 1,058 mechanically enumerated); the CI workflow (`.github/workflows/fusion-category-ci.yml`, with the completeness gate covering `lake build`'s import-closure blind spot); and the clean-build kit (`tools/clean-build.sh`). The batch's findings that changed this document: the $(\mathbb{Z}/3)^2$ $\zeta^{a_1 b_2 c_1}$ family is a coboundary (the third-witness Remark now quotes the corrected family); the instance table's $\mathbb{R}$ entries for the two $\mathrm{Vec}_{\mathbb{Z}/2}$ data (they are $\mathbb{Q}$-valued, so the $\mathbb{R}$ originals are instantiated directly by three of the six); `eno823_exact` tied to the name Task 1.1 freezes; WP0's "no network" criterion restated (the Mathlib cache server is a hard dependency of a fresh machine); the inventory recomputed (31 content files, 8,135 lines, 1,058 audited declarations). The clean build is reproduced; the audit is mechanical and green; the self-test is 13/13 (a prime-name fixture joined it after the landing itself found that Lean identifiers may contain primes, which the gate's recogniser and name extraction now handle greedily). The batch's one open item closed with the follow-up run: the honest-twist witness `Examples/Z2SqHonest.lean` (Gate 1's nonconstant-pivotal witness, the positive instance of the corrected target, and the forcing theorems showing `J_match` is a consequence rather than extra data on that datum).